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# END-TO-END LEARNING TO PREDICT ARCHEOLOGICAL POTENTIAL

Integrating Deep Learning for Improved  
Archeological Site Prediction

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August 2024

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**Science and Bio-Engineering Sciences**



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Proefschrift ingediend met het oog op het behalen van de graad van Master of Science in de Ingenieurswetenschappen: Computerwetenschappen

# END-TO-END LEARNING VOOR HET VOORSPELLEN VAN ARCHEOLOGISCH POTENTIEEL

Deep learning integratie voor verbeterde  
voorspelling van potentiële  
archeologische sites

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# Abstract

Archaeologists uncover guiding factors in human history, providing deep knowledge about the evolution of societies through the study of artefacts found at archaeological sites. The research and documentation of these archaeological sites is often time-intensive and costly. To assist archaeologists in this work, predictive models are used to identify potential sites based on local geographical features. The models classically use feature extraction on which a regression analysis follows. To minimise the possible introduction of biases in feature extraction, an end-to-end method is proposed in this work. Similarly, naive Bayes is introduced as a possible tool for preliminary efficient analysis of archaeological potential while eliminating the need for manual parameter selection, further reducing possible biases. Both are trained on geospatial information gathered by remote sensing techniques and annotated features such as existing archaeological sites, distance to historical road networks and distance to urban centres. The methods were compared and evaluated against an existing technique in the Sagalassos survey area, situated in the western-Taurus mountain range in Anatolia. In the applied experimental setting, naive Bayes demonstrates to be superior against the end-to-end approach and existing method. Multiple complexities, introduced by the lack a general benchmark, are revealed in the analysis and prevent global claims. This work takes the first steps towards integration of much promising neural networks in the research field and can be used as a basis for further research.

# Samenvatting

Archeologen onderzoeken drijvende factoren in de menselijke geschiedenis en verschaffen diepgaande kennis over de evolutie van samenlevingen door artefacten te bestuderen gevonden bij archeologische sites. Het onderzoek en documentatie van deze archeologische vindplaatsen is vaak tijds- en resourceintensief. Om archeologen bij dit werk te ondersteunen, worden predictieve modellen gebruikt om potentiële vindplaatsen te identificeren op basis van lokale geografische kenmerken. De modellen maken klassiek gebruik van feature-extractie waarop een regressieanalyse volgt. Om de mogelijke introductie van vooroordelen bij de extractie van kenmerken te minimaliseren, wordt in dit werk een end-to-end methode voorgesteld. Op vergelijkbare wijze wordt naïeve Bayes geïntroduceerd als een mogelijk hulpmiddel in het uitvoeren van efficiënte inleidende analyse van archeologisch potentieel, waarbij de noodzaak voor handmatige parameters wordt geëlimineerd waardoor mogelijke vooroordelen worden beperkt. Beide methoden worden getraind op basis van geografische informatie verzameld door middel van afstandswaarnemingen en geannoteerde kenmerken zoals bestaande archeologische vindplaatsen, afstand tot historische wegennetwerken en afstand tot stedelijke centra. Een vergelijkende studie evalueert het potentieel van de geïntroduceerde methoden in het Sagalassos onderzoeksgebied, gelegen in het westelijke Taurusgebergte in Anatolië. In de toegepaste experimentele setting blijkt naïeve Bayes superieur ten opzichte van de end-to-end benadering en de bestaande methode. Meerdere complexiteiten, geïntroduceerd door gebrek aan een algemene referentiekader, komen aan het licht in de analyse en verhinderen globale conclusies.

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# Chapter 1

## Introduction

There is a growing amount of attention from governments to include archaeological sites in processing urbanisation. Designing infrastructure for tomorrow's society increasingly uses and incorporates historical and archaeological values from the past [27]. Archaeologists are trying to map out the archaeological legacy to realise a source of information and inspiration for future generations. Due to lack of resources, it is impossible to comb every square metre in the search for historical occupancy indications. There is a need for maps showing not only known existing archaeological sites but also indications of regional archaeological potential. This potential of historical land-use, in the literature and this dissertation, is called archaeological potential and is defined as: "The relative suitability of different land parcels within a confined region for human occupation" [14, p. 3371]. Archaeological predictive modelling (APM) is used to predict the potential of archaeological sites in a landscape. Studies on this topic began in the early 1980s [27]. Modern applications take advantage of rapid advances in computational capabilities in statistics, artificial intelligence and machine learning [72, 76, 24]. A fundamental assumption is that ancient human land use is guided by human decisions influenced by patterns in natural environmental variables [24]. An additional assumption is that these variables are still present in the modern landscape and can be collected by current Geographic Information System (GIS) techniques. In essence, a predictive model tries to infer a prediction for new data points using a quantitative analysis of data points consisting of environmental factors and a site absence/presence label. Models are selective abstractions that inevitably omit much of the real world's complexity. The data or prior knowledge represented in a model are chosen based on what the creator deems significant, according to their interests and the problem at hand. This concept is crucial because it indicates that no model is entirely objective, whether constructed inductively or deductively. Consequently, all models inherently reflect the subjectivity and bias of their creators [57] and so far there is no consensus on standard methods or techniques to create archaeological predictive models. In this dissertation, we probe a contribution to archaeological predictive modelling research field by developing an end-to-end model using a deep learning semantic segmentation architecture applied in the Sagalassos area located in Turkey. This approach aims to reduce the subjectivity inherent in traditional modelling techniques, offering more nuanced predictive surfaces. In addition, naive Bayes is introduced into the APM field as a tool to make efficient baseline predictions.



## 1.1 Problem Orientation

Modern applications of archaeological predictive modelling methods such as logistic regression, weight-of-evidence and random forests have shown interesting results [23, 76, 45] but are limited by their simplicity. In logistic regression, for example, a linear relationship is expected between independent variables and log odds of the dependent variable [33]. This expectation undermines the inclusion of non-linear patterns in the data. Research shows susceptibility to under-fitting and low accuracy when logistic regression is used for APM [76]. Weight-of-evidence and random forests, discussed further in related works, show weak performance for modelling complex relationships [24, 48]. Deep neural networks (DNN) offer a common alternative. DNNs' layered architecture consisting of inter-connected neurons can represent data at different levels of abstraction and model complex, non-linear relationships [26]. A specialised form of DNNs, convolutional neural networks (CNNs) have been the foundation for brilliant achievements and have become one of the most representative neural networks in the field of deep learning [49]. CNNs also show great potential in modelling spatial patterns in collected GIS data [9, 68] but the literature lacks applications of this methodology in APM. Additionally, naive Bayes is introduced as a response to the need for a simplistic baseline model. Using naive Bayes avoids the need for selecting specific, often region-dependent, parameters to optimise predictions. Manual parameter selection introduces susceptibility to Bayes and makes it difficult to compare applications in different regions.

## 1.2 Scope

In this thesis, dedicated sections provide the necessary background in machine learning as well as archaeological data collection. In the related work section, current methods in APM are discussed followed by a critical analysis. One such method, the locally adaptive model for archaeological potential (LAMAP), shows positive indications of archaeological potential after analysis in the study area. However, as will be cited, these, probability theory-based methods come at a high computational cost. This dissertation introduces a well-known machine learning technique naive Bayes as an attractive alternative as a fast tool for archaeological potential research. However, LAMAP, naive Bayes and other current methods lack capabilities to account for complex natural environmental dependencies and processes. A specialised deep neural network architecture, semantic segmentation is proposed as solution. The performance of LAMAP, naive Bayes and semantic segmentation methods in predicting archaeological potential are compared in a cross-validation study with ground-truth known archaeological sites and known non-archaeological sites.

# Chapter 2

## Related work

### 2.1 Overview

This chapter studies the necessary background and related works on which the proposed contribution of semantic segmentation and naive Bayes to APM is motivated. It first describes APM as a research area from its first to some modern methods and how machine learning fits into it. Next, LAMAP and the criticisms on classical APM methods for which it provides solutions is discussed. As will be shown, LAMAP's solutions come with a high computational cost. This specifically provides the motivation for the later introduction of the naive Bayes method. Also, LAMAP will show lack of capacity to model complex phenomena. To answer this, semantic segmentation is introduced starting from its origins in artificial neural networks.

### 2.2 Archaeological Predictive Modelling

Machine learning is increasingly finding its way into archaeology as a tool in analysing spatial, cultural, textual, natural and artistic data with the major bottleneck being data scarcity. Gathering sufficient archaeological data to render ML applicable is challenging. Vulnerability to hidden complexity, systematic bias and high cost of validation are some of the consequences. Research in machine learning is advancing, which makes applicability in archaeology increasingly feasible [8]. However, in this research, predicting archaeological potential, has a major advantage over other applications within archaeology. Indeed, in this context, geospatial data collected by lidar, satellite and aerial imagery on local, regional and national scales can be used. This data will form the foundation in predicting archaeological potential. It is important to underline that in this study, no exact sites locations but archaeological potential is predicted. Archaeological potential examines the contribution of various natural and/or cultural variables such as slope, elevation and distance from cultural centre to useful indicators that point to potential archaeological sites [8]. In the following sections, some classical methods are first discussed which can be roughly divided into two groups. On the one hand, a model can be based on a theoretical perspective, this model is then deductive or theory-driven. On the other hand, only observations can be used which provides an inductive or data-driven model. Although theoretically pure implementations are possible in either of these groups, most archaeologists do not make a distinction as observations are often collected in a theoretical context and theory is supported by observations [79]. This gives rise to the introduction of machine learning in archaeological research. Some of the most popular methods are presented and analyzed.

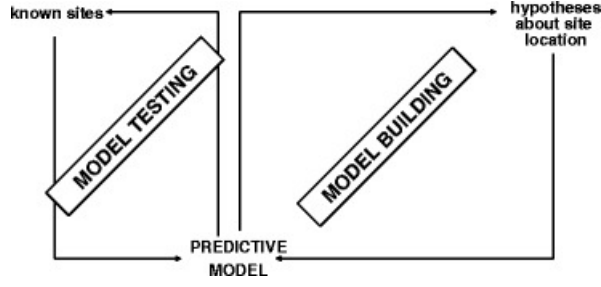


Figure 2.1: The procedure used for deductive predictive modelling. Building the model is theory-driven and consists of educated guesses. Known sites are only used for testing purposes, not to build the model [75].

### 2.2.1 Rule-based approach

This approach is based on a set of rules which define the conditions under which a given area has archaeological potential. The deduction of these rules can originate from either a theoretical or an observational approach.

#### Deductive Rule-Based approach

This approach is less frequent but still has use-cases in archaeological land evaluation [39, 38]. Archaeologists can draw on expertise and experience to create rules that frame archaeological potential. These rules could for example apply to natural non-archaeological characteristics in the landscape which, according to archaeological study, correlate with the presence or absence of an archaeological site. These characteristics are, for example, slope, distance to flowing water, soil type and aspect. The rules are used in a multivariate discriminant function as in equation 2 where the characteristic are the variables,  $M$  the set of sites and  $M'$  the set of non-sites [79].

$$M' = (Slope < 10) \cap (Distance < 1km) \cap (Soil = A) \cap (Aspect = South) \quad (2.1)$$

$$M = (Slope \geq 10) \cap (Distance \geq 1km) \cap (Soil \neq A) \cap (Aspect \neq South) \quad (2.2)$$

In Equations 2.1 and 2.2 archaeological knowledge is used to define fitting hyper-parameters. This approach is straightforward and can fall back on theoretical knowledge based not only on proper geological but also ecosystemic structures and cultural behaviour characteristic observed by experts. Settlement locations are not only driven by geological characters but also by the relationship to other settlements in the form of competition, cooperation and communication. The availability of communities and resources thus play a key role in site location [37]. All this data can be encompassed in a complex multivariate discriminant function but it is clear that biases are easily introduced via selecting characteristics and hyper-parameters.

#### Inductive Rule-Based approach

In an inductive approach, rules are based on the characteristic of existing data about known sites. When predicting ancient trade routes from a particular era, sites where parts of these routes are found can shape the rules instead of defining the rules deductively [79]. Of the surveyed sites, for example, their distance to certain geological features such as waterways and fertile land are recorded. The continuous nature of a characteristic like distance to water can be divided into

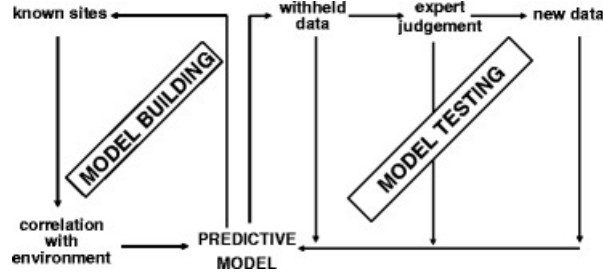


Figure 2.2: The procedure used for inductive predictive modelling. Known sites are used to derive information for the predictive model. The model is tested using statistical methods or expert systems [75].

three classes: adjacent, near and far. The expected number of sites for each terrain type is calculated and weighted to account for biases in the distribution of archaeological sites. The proportion of sites in a certain class, times its quantity, provides the expected number of sites. Likewise, different variables can be weighted to represent preferences. A predictive model can thus be based on terrain types that were favoured by ancient traders. An example is given in Equation 2.1. A significance test can also be included to ensure that the characteristic used affect the distribution of archaeological sites. Weights are calculated for each variable chosen by an expert. When these are combined in an equation, it is called the weights of evidence method. The equation is presented as follows:

$$P(\text{Archeology}|\text{Evidence}) = P(\text{Archeology}) * \text{criteria1} * \text{criteria2} * \text{criteria3} * \dots$$

Here  $P(\text{Archeology})$  is the prior and  $P(\text{Archeology}|\text{Evidence})$  the posterior probability of finding an archaeological site given some landscape characteristics of the area. Evidence features such as, for example, flat terrain or close distance to water will introduce weights which have positive impact on the archaeological probability.

Table 2.1: Distribution of Archaeological Sites by Distance to Water. One method of calculating ‘weights’ for an input variable [79].

| Distance to Water  | 0–499m | 500–999m | 1000m+ | Total |
|--------------------|--------|----------|--------|-------|
| Proportion of area | 60%    | 35%      | 5%     | 100%  |
| Number of sites    | 15     | 45       | 10     | 70    |
| Expected           | 42     | 24.5     | 10     |       |
| Weighting          | 0.35   | 1.8      | 0.3    |       |

### 2.2.2 Machine Learning

Machine learning [57] is a discipline within the field of artificial intelligence focused on developing algorithms that improve their performance in solving tasks by gaining experience from data. The dichotomy between deductive and inductive approaches has little meaning in this context because data underpins the learning process yet the choice of data points and its attributes is guided by the prior knowledge and biases of its creator. Three major subgroups within machine learning can be distinguished.

- In supervised learning, models are trained on labeled data, where each data point is paired with a corresponding label. This allows the model to learn a direct mapping from input to output, making it well-suited for tasks like classification, where the goal is to assign categories to inputs, and regression, where the aim is to predict continuous values.
- In unsupervised learning, unlabelled data is used. Algorithms in these subgroups try to recognise underlying patterns. Methods in this group are clustering algorithms and dimensionality reduction techniques.
- Reinforcement learning, on the other hand, involves training a model through interactions with an environment. Instead of learning from static data, the model learns by taking a sequence of actions and receiving feedback in the form of rewards for desirable actions or penalties for undesirable ones.

In the context of archaeological predictive modelling, the primary focus is on supervised learning. This approach relies on training data that includes labels indicating the presence or absence of archaeological sites. The input data consists of spatial landscape features derived from remote sensing, providing a pixel map over an area, with each pixel representing a specific measurement at a given resolution, such as 30m x 30m. Ideally, each pixel would be accompanied by a ground truth label which is either positive if the pixel corresponds to a known site location, or negative if corresponds to a known non-site location. In practice, however, the resulting known non-site labels have limited reliability because the lack of artifacts on a surface is not necessarily caused by lack of historical occupation but could be due to natural phenomena such as erosion that erased the evidence. Finding archaeological artifacts indicates a site but not finding them does not necessarily indicate a non-site. A common practice in APM is to make a random selection of negative data points. This practice is based on the assumption that sites are rare and therefore only occupy a very small percentage of the survey area [77]. Machine learning makes good use of the large amount of available GIS data and has already provided many positive results in diverse environments [11, 15, 29, 73]. The following is a discussion of two of the most commonly applied machine learning techniques in predictive archaeological modelling.

### Random Forests

Random forest (RF) is a popular ensemble technique in machine learning based on decision trees which have shown positive results in predictive modelling [84]. Decision trees [57] model decisions and their consequences according to a tree architecture. Each node in this tree represents a decision that classifies data points according to the value of a particular feature in the data. Each end-node (leaf) is a final decision or classification. The tree is built recursively by progressively classifying the data. Starting with the entire dataset, a feature is determined in the root node that best subdivides the data points into classes according to the chosen metrics such as Gini impurity or entropy (information gain). For each subclass, this process is repeated up to a certain stop criteria such as reaching a homogeneous set or the maximum tree depth. The hierarchical tree structure is easy to interpret and is therefore popular when explainability is needed. However, decision trees are prone to over-fitting. Over-fitting happens when the model includes not only underlying patterns but also noise in the training data itself. As a result, the model will perform well on the data it was trained on but not well on new data. This consequence is called poor generalisation in machine learning. Random forests overcome this problem by using an ensemble of individual decision trees. Each tree in the ensemble is trained using a subset of the training data, generated through a technique called bootstrapping, where multiple new datasets are created by randomly sampling the original dataset with replacement.

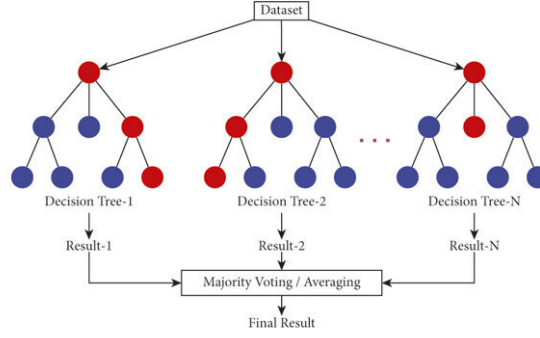


Figure 2.3: Illustration of random forest trees [42]

The output when predicting a continuous variable becomes the average over the individual trees and in the majority vote over the trees in categorical predictions. Furthermore, random feature selection is also often used where each split in the tree construction considers a random sub-sample of the available variables. The introduction of randomness in bootstrapping and feature selection provides better resistance to noise in the data and great generalizability capabilities.

### Logistic Regression

Besides weights of evidence, logistic regression models are a prominently used method which has its origins in linear regression. While the approaches discussed in the previous section serve as adequate methods for generating predictions, they fail to answer a number of crucial questions, including the degree of influence individual predictors exert on the model, methodologies for predicting ordinal or continuous archaeological variables such as site or artefact density and the overall fit of the model to the observed distribution of sites [79, 80]. This is where regression analysis can be used. The output of this model represents the probability between 0 and 1 of an event (such as a positive site) given environmental parameters. These parameters are independent variables for which the model examines their relationship to the dependent variable (1 for a site and 0 for a non-site) and calculates the dependent probability of a positive site for each new data point. To better understand logistic regression, it helps to look at linear regression first [57]. Linear regression models a linear relationship between independent variables and a continuous dependent variable. The model does this by defining a line for which the distance between the available data points and its predictions is minimised. The line is represented as the function  $\hat{f}(x) = w_0 + w_1 a_1(x) + \dots + w_n a_n(x)$ , where  $a_i(x)$  denotes the  $i$ th input element,  $w_0$  the intercept and  $\langle w_1 \dots w_n \rangle$  the learned weights for each feature. Because the predictions in linear regression are not closed within the  $[0, 1]$  interval, it is not applicable for predicting probabilities. For this purpose, logistic regression is introduced. Instead of a real valued, the output in logistic regression can be interpreted as probability by using a sigmoid function. This function is applied to the same linear combination of features as invoked in linear regression. The models output  $o$  is represented as  $o = \sigma(\vec{w} \cdot \vec{x})$ , where  $\sigma$  the sigmoid function:

$$\sigma(y) = \frac{1}{1 + e^{-y}}$$

In essence, logistic regression transforms the classical linear regression model so that it is capable of binary classification by using the sigmoid function. It is interesting to note that this function will also play an important role in the neural networks introduced in section 2.5.2

### 2.2.3 Critique on Predictive Modelling

Much criticism is rooted in the introduction of biases in modern methods that use large amounts of data. Biases can be introduced in through, for example, poor GIS application or biased sample selection [81]. As also cited in 2.2.2, labelled data consisting of both known sites and known non-sites is needed in supervised machine learning. In practice, non-sites are often selected randomly. Despite the assumption that site occurrence is rare, the model will learn to distinguish known sites from unknown sites rather than known sites from known negative ones. Weight of evidence and logistic regression are commonly used but subject to much criticism [13]. For example, although the weights of evidence method is conceptually and statistically very simple, it does not take into account, possible correlations between input variables [79]. Research has already shown that these correlations have a significant impact on site locations [19]. Although logistic regression is non-parametric against input variables and thus makes no assumption about the form of relationship between inputs and outcomes, logistic distributions are still used to arrive at a probability distribution on a binary outcome. The distribution provides a monotonic relationship so that as the input variable increases, the probability of the outcome being in a particular class also increases. If theory predicts the probability distribution forms a two-peaked pattern for a certain predictor variable and thus is not accurately modifiable with a monotonic function, the model must be split or the variable divided into two groups [81]. In summary, several problems with both weights of evidence and logit regression are found. Problems arise from violating statistical assumptions, modelling complex relationships using single variables and the dependence on often scarce non-site data. The locally-adaptive model of archaeological potential (LAMAP) model [13] attempts to address these problems.

## 2.3 Locally-Adaptive Model of Archaeological Potential

It is fundamental to distinguish between modelling known site locations to predict undiscovered sites according to a probability using data derived from known sites and the concept of archaeological potential. The latter is central to LAMAP. The following theoretical and practical explanations of LAMAP and its implementation are based on papers by Carleton [13, 14] and Willet. P. [81] where LAMAP is applied to the Sagalassos study region that is also studied in this work.

### 2.3.1 Archaeological Potential and Theory Behind LAMAP

LAMAP models the locations for certain human behaviors and the spatial distribution of parameters influencing those patterns that are observable in the dataset of known-sites i.e., the existing archaeological record of the area of interest. The objective is to model the suitability of a given area to support a human activity during a specific time period using spatial distributions of parameters influencing this documented in a dataset of known-sites. Ancient civilisations based settlement choices not only on heuristic observations of the landscape such as elevation, slope and distance to waterways but also on cultural precedents and proximity bias due to the high cost of finding a suitable site. A final step in the LAMAP method is to scale up the result to the highest and lowest value because predictions are relative to other nearby locations rather than describing a suitable location. LAMAP makes distributions of know-sites and compares them with 'unknown' locations to predict potential. So in LAMAP, contrary to logistic regression and weight of evidence models, there is no need for non-site data. This method is a hybrid of a data-driven and theory-driven approach because inductive methods take care of

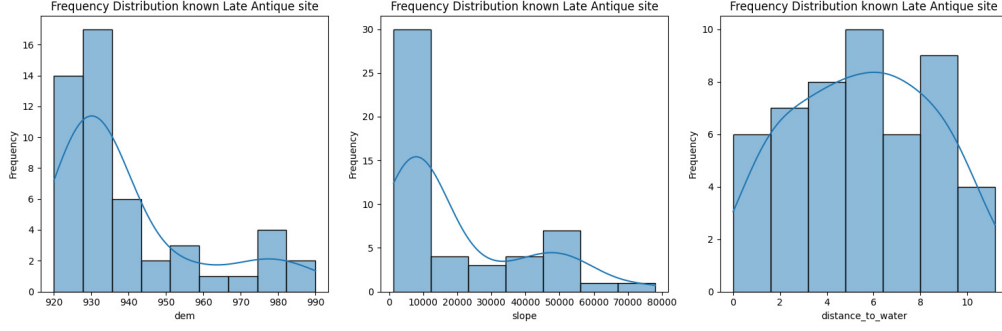


Figure 2.4: Frequency distribution plots of elevation, slope, and distance to water for the 50 nearest points of a known site in the Late Antique period in the dataset.

weighting the parameters and the deductive selection of determinant variables by the researcher is theory-based.

### 2.3.2 Underlying Calculations of LAMAP

Raster maps containing relevant landscape data such as height, slope and distance from each pixel to a relevant feature such as a river form the input of the LAMAP method. In LAMAP an archaeological site example comprises multiple geospatial cells in its neighbourhood so all landscape variables can be described as a multidimensional joint distribution. Computing a predictive area using LAMAP is essentially comparing each pixel, which are characterised by the same landscape variables, to each known-site distribution. In this way the answer to the question: "What is the probability that this target pixel occurs in the known-site dataset?" is calculated.

#### Step 1: Known-site Empirical Frequency Distributions

Each known site location is modelled by its distribution of landscape variables which are used to calculate the probability,  $y$ , by which a given pixel occurs in this region. Consider the elevation gathered from a digital elevation model (DEM). A certain radius is chosen around the known site location and the known site elevation distribution is obtained by counting the occurrences of each elevation in the area of interest bounded by the radius.

Assume  $(v_1, v_2, \dots, v_n)$  all elevation values within a certain defined radius around a known archaeological site coordinate. The frequency distribution  $\widehat{F}_{k,V}$  of the height values  $V$  in the radius around site  $k$  is defined as:

$$\widehat{F}_{k,V}(v) = \sum_{i=1}^n I(v_i),$$

where  $I$  is an indicator function:

$$I = \begin{cases} 1 & v_i = v \\ 0 & \text{otherwise} \end{cases}$$

An example is visualised in 2.4. This process is repeated for each land axis feature for each known set, resulting in a set of frequency distribution functions for each known-site.



### Step 2: Probability of the Target Pixel Given One Known-Site

The goal of this step is to predict the probability of finding a particular pixel similar to the target pixel in a known-site area. Take a landscape feature  $V$  of a known-site  $k$ . The key question is “What is the probability of encountering a pixel in the known-site area with a particular value for the landscape variable?”. To compute this probability, each target value for  $V$ , called  $v_{tp}$  is compared to the frequency distribution  $\widehat{F}_{k,V}(v)$  for this known site  $k$ . The probability  $\lambda_{v_{tp},k}$  of finding  $v_{tp}$  in the area of  $k$  is the ratio between  $\widehat{F}_{k,V}(v_{tp})$  and  $\widehat{F}_{k,V}(v)$  for all features  $V$  in the environment of  $k$ .

$$\lambda_{v_{tp},k} = \frac{\widehat{F}_{k,V}(v_{tp})}{\sum_{i=1}^n \widehat{F}_{k,V}(v_i)},$$

where  $v_i$  is a possible value of  $n$  possibilities of  $V$  in  $\widehat{F}_{k,V}(v)$

However, there is a need to provide a small margin around the target. Using the formula described above the exact probability is calculated but the probability of an exact outcome in a continuous distribution is 0. Also, from anthropological perspective it is plausible that people do not approach landscape variables with great precision in choosing area occupancy. For example, instead of calculating the probability that a pixel with an exact height of 200m occurs in a known site area, a range around this value is used for example  $200 \pm 5$ . The interval is written as  $v_{tp} \pm h$ . The empirical frequency distribution is transformed into a cumulative distribution function (CDF) defined as:

$$F_{CDF,k,v}(v) = \frac{1}{n} \sum_{i=1}^n I_{CDF,k,V}(v_i)$$

Where  $I$  is the interval function:

$$I_{CDF,k,V}(v_i) = \begin{cases} 1 & v_i \leq v \\ 0 & \text{otherwise} \end{cases}$$

This CDF function, visualized in Figure 2.5, gives the probability of finding a pixel area  $k$  with an attribute value less than or equal to the given  $v$  value. This probability is calculated for both the upper and lower bounds, and the following formula gives the probability of the value occurring in the interval  $v_{tp} \pm h$ :

$$\lambda_{v_{tp} \pm h,k} = F_{CDF,\widehat{F},k,v}(v_{tp} + h) - F_{CDF,\widehat{F},k,v}(v_{tp} - h)$$

This method must now be applied for each landscape variable. The result is a list of probabilities  $p_{tp,k}$  representing the probability of finding a pixel with height  $v_{elev} \pm h_{elev}$ , a slope of  $v_{slope} \pm h_{slope}$  and a distance to water of  $v_{dist} \pm h_{dist}$  and so on. The combination to one value of this list implies an intersection of probabilities. It can be assumed that the individual probabilities are independent because values for one variable do not affect values of other variables. This allows the intersection to be calculated using multiplication. The formula is given as:

$$\bigcap p_{tp,k} = \prod_{i=1}^n p_{tp,k,V_i}$$

The final probability represents the probability of finding a pixel like the target pixel,  $tp$ , near known site  $k$  taking into account all landscape variables. The method so far gives us the probability versus one known site. The calculation is re-iterated for each known site so that a new list  $P_{tp,k}$  is obtained.

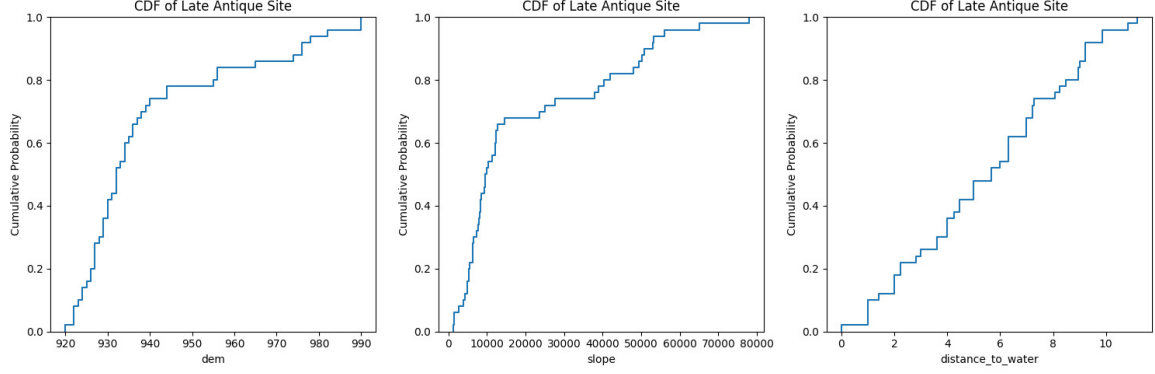


Figure 2.5: CDF plots of dem, slope, and distance to water for the 50 nearest points of a known site in the Late Antique period in the dataset.

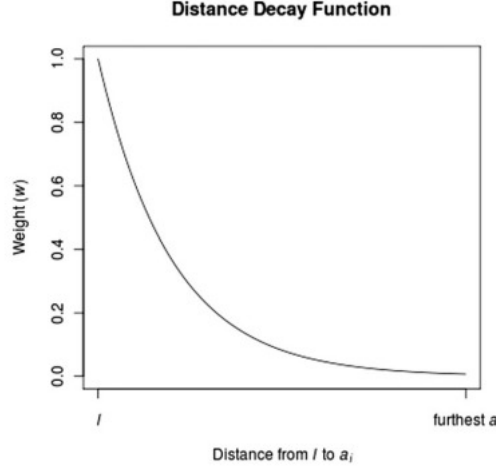


Figure 2.6: The optional distance-weighting function

### Step 3: Distance weighting

Intuitively, neighboring sites will be more similar to each other than distant sites. We expect the ecologies of nearby sites to be more similar to the target pixel ecology than more distant sites. It is also expected that given the strong dependence on geography and geology people will base land use decisions on decisions made by people in nearby areas and to a lesser extent on those made in distant areas subject to very different geography. Based on this assumption, a smooth exponential decay function is proposed for weighting the probabilities in  $P_{tp,k}$  according to their distance from the target pixel.

### Step 4: Total Probability Given All Known-Sites

The weighted list of probabilities  $P_{tp,k}$  should now be combined into a value showing the archaeological potential of a pixel. The different probabilities cannot be combined by simple multiplication as they are not mutually exclusive. One value represents the probability of finding the

target pixel for one site but a pixel like the target pixel can occur in the area of several sites. In probability-theory, this means that the intersection of the two events is not empty and the probability of the intersection must also be taken into account. To solve this, the Law of Total Probability for non-mutually exclusive events with the Inclusion-Exclusion Principle is used:

$$\begin{aligned}
 P(\cup_{i=1}^n A_i) &= \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \\
 &\quad + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n-1} P(\cap_{i=1}^n A_i) \\
 &= \sum_{k=1}^n ((-1)^{k-1}) \sum_{I \subseteq \{1,2,\dots,n\}, |I|=k} P(\cap_{i \in I} A_i)
 \end{aligned}$$

This combination gives the final LAMAP prediction. The 4 steps are performed on each individual pixel within the study area.

### 2.3.3 Analysis of Methodological Strengths and Pitfalls

In the LAMAP model, the absence of parameterization for natural spatial phenomena avoids the necessity for complex parametric models, which are essential for accurately representing these phenomena [13]. This enables the risk of oversimplifying the spatial dependencies and patterns [56]. Consequently, the model may lack the sophistication needed to capture subtle spatial dependencies and variations [21]. Each site is considered separately and weighted according to its distance from the target point. This weighting introduces the bias that short-lived known sites areas have greater predictive power [81]. This may ignore the broader context and interactions between different site types. There is no need for non-site data in this method which is very attractive for lower cost research projects that do not have the resources to collect this data. Not including non-site locations can, however, fail to generalize to areas with different unseen spatial features. The model requires multiple decisions which provides flexibility depending on the area of study or particular analytical goals. Examples of choices include: the size of the interval used in the ECDF equations and the shape of the distance decay function. This flexibility in decision-making regarding model parameters, sample areas, and weighting functions can however introduce subjectivity and inconsistency, potentially leading to varying results across different studies.

### 2.3.4 Analysis of Time Complexity

LAMAP facilitates a predictive method applicable in an environment with large landscape variation over relatively small areas such as in the Sagalassos Study Area. This strength comes from its ability to consider each known site as independent and give nearby sites greater influence. However, this comes at a potentially huge computational cost when computing the total probability given all known sites. The technique used, the inclusion exclusion principle (IEP) [16], was devised in 1718 by Abraham de Moivre [67]. IEP states that the total probability of a set of events equals the sum of the probability of each individual event minus the probability of the intersection of each compound event plus the probability of the intersection of all events. The total probability for 2 events is described as:

$$\begin{aligned} Pr\{A \cup B\} &= Pr\{A\} + Pr\{B\} \\ &\quad - Pr\{A \cap B\} \end{aligned}$$

For 3 events, this becomes:

$$\begin{aligned} Pr\{A \cup B \cup C\} &= Pr\{A\} + Pr\{B\} + Pr\{C\} \\ &\quad - Pr\{A \cap B\} - Pr\{B \cap C\} \\ &\quad - Pr\{A \cap C\} + Pr\{A \cap B \cap C\} \end{aligned}$$

The IEP is equivalent to performing a power set expansion which is the process of generating the set of all possible subsets from a given set. This process causes a time and space complexity of  $O(2^n)$  where  $n$  is the number of unique events. In the context of applying LAMAP,  $n$  is the number of unique known sites. In complexity analysis, only the most expensive factor is taken into account, but it should be emphasized that the IEP is applied for each pixel in a survey area. For example, the Sagalassos survey area covers 1450 km<sup>2</sup> which for the application of LAMAP is divided into 50x50m cells resulting in 580,000 target pixels. Creating predictive surfaces with LAMAP requires specialized hardware such as supercomputers to handle the astronomical scale of computations. Such hardware has limited access and requires the necessary knowledge to use. This adds another layer of complexity to the application which makes it difficult and intensive to apply LAMAP iteratively with different parameter values. However, even with specialised hardware, executing a process with  $O(2^n)$  is not scalable to increasing datasets due to exponential growth in the computational resources needed. There is a need for a scalable methodology accessible with a minimum of computational resources so that researchers can focus on the subtleties and make easy reiterations of the model.

## 2.4 Introducing naive Bayes for Baseline Predictions

In essence, LAMAP has much in common with a well-known technique, the naive Bayes classifier, from machine learning [57]. Bayesian reasoning supports a probabilistic approach to inference based on the assumption that the quantities of interest are modular under probability distributions and optimal decisions can be learned by reasoning about these distributions and observed data. Bayesian learning algorithms such as naive Bayes explicitly compute the probability for hypotheses which is of great advantage for certain types of problems. In these settings, another major advantage, especially over LAMAP, is that there is no need for the problem of parameter selection introduced in Section 2.3.3. Naive Bayes does not have any mechanism to adapt to, for example, specific study areas. This allows easy comparison of results in different environments. As will be discussed in this section, naive Bayes also provides answers to the high computational needs of LAMAP.

### 2.4.1 Underlying Calculations of naive Bayes

The learning environment is described as the context where each data point  $x$  is defined by a conjunction of attributes and where the target function  $f(A)$  can take any value from a finite set. A training set is composed of tuple attribute values  $(a_1, a_2, \dots, a_n)$  labeled by the  $f(A)$  function. In naive Bayes classification, a new data point with associated attribute values  $\langle a_1, a_2, \dots, a_n \rangle$  is classified with the most probable target value  $v_{MAP}$  given as:

$$v_{MAP} = \underset{v_j \in V}{argmax} P(v_j | a_1, a_2, \dots, a_n) \quad (2.3)$$

Using Bayes theorem this becomes:

$$\begin{aligned} v_{MAP} &= \underset{v_j \in V}{argmax} \frac{P(a_1, a_2, \dots, a_n | v_j) P(v_j)}{P(a_1, a_2, \dots, a_n)} \\ &= \underset{v_j \in V}{argmax} P(a_1, a_2, \dots, a_n | v_j) P(v_j) \end{aligned}$$

The probability  $P(v_j)$  is easily estimated by counting the frequently with which  $v_j$  occurs in the available data. To calculate the  $P(a_1, a_2, \dots, a_n | v_j)$  term, the classifier is based on the assumption that the attribute values are conditionally independent given the target value. The probability of the conjunction  $a_1, a_2, \dots, a_n$  is the product of the probability of the individual attributes given the label. Incorporating this approach into the formula gives the basis of the naive Bayes classifier:

$$v_{NB} = \underset{v_j \in V}{argmax} P(v_j) \prod_i P(a_i | v_j) \quad (2.4)$$

The naive Bayes method first applies a learning step in which the  $P(v_i)$  and  $P(a_i | v_j)$  terms are estimated using the available data. The preceding Equation 2.4 can now be used to classify new data points. It should be clearly emphasized that while the assumption of conditional feature independence given the class label is often an oversimplification, it allows for easy model fitting and still performs well in practice. As also discussed in section 2.3.3, a different approach is needed to deal with calculating probabilities for continuous data. A typical assumption made in this case is that the continuous values correlating with each class follow a Gaussian distribution. The data sets are divided by class for which the mean and standard deviation are calculated. The following formula estimates the probability in a continuous case:

$$P(a|v) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}} \quad (2.5)$$

Where  $x$  = variable,  $c$  = class,  $\mu$  = mean,  $\sigma$  = standard deviation. When the Gaussian distribution assumption is used it, is referred to as Gaussian naive Bayes. In the LAMAP section we discussed how this method deals with the problem of calculating probabilities on continuous data. Instead of using an exact attribute value, an interval is taken around these values for which the total probability can be calculated elegantly using the CDF. In Gaussian naive Bayes, the Gaussian distribution assumption is made to overcome this problem.

### 2.4.2 Application of naive Bayes in APM

In order to create an easily re-iterable baseline model, combining probabilities according to the IEP should be avoided. The LAMAP method doesn't explicitly model the definition of an archaeological site. Each known site is modelled independently by using empirical data points in the vicinity of each site to construct a CDF for each landscape variable. By comparing each new data point to each of these learned definitions, giving more weight to spatially close examples, the archaeological potential of this new point is estimated. In this way the definition of an archaeological site is locally adaptive with as drawback the need for the expensive IEP to combine prediction from all independent sites. Naive Bayes is applied to 1 class: 'known sites'.

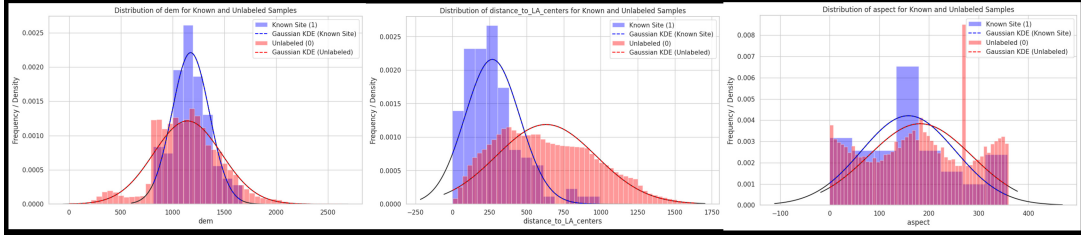


Figure 2.7: Distributions of key features for both labeled and unlabeled data points: the elevation feature (left), the distance to Late Antique city centers (middle), and the aspect feature (right). It should be noted that only known positives are utilized in estimating the potential for archaeological sites.

Instead of considering each site separately, all are considered equal, all as positive data points. Philosophically, this just teaches what a general positive site looks like based only on the true positive data points. The definition of a positive site is contained in a probability density function. One for each landscape variable with the assumption that they follow a normal distribution. The attributes of a new data point are compared to their corresponding distributions and since the independence of landscape variables is assumed to be true, the comparisons are combined by multiplication to elegantly obtain by which probability the new point belongs to the positive sites. In other words, how much the new point conforms to the definition of a positive site is calculated. How much does this point resemble an archaeological site. What is the archaeological potential of this point. The naive Bayes method is very efficient compared to the LAMAP method, it has a run time complexity of  $O(d * c)$  where  $d$  is the number of landscape attributes of the data point and  $c$  is the number of possible classes which is one in this case.

## 2.5 Semantic segmentation for End-to-End Predictions

Semantic segmentation is the task within computer vision to label each pixel within an input image with a predicted category to which the pixel belongs. Opposed to this are tasks such as image classification where one label is predicted for an entire image and object detection where the location and identity of an object within an image is predicted. With great success in various computer vision applications, semantic segmentation has also found its way into remote sensing image analysis [85]. For example, its pixel-wise classification facilitates the identification of various objects and structures in urban spaces [68] and environmental monitoring [9]. The backbone of semantic segmentation are Convolutional Neural Networks (CNNs) which are a form of Artificial Neural Networks (ANNs) specializing in grid-like data such as images. The following sections discuss its building blocks as well as possible CNN architectures which enable semantic segmentation. It will first be described how the neural network-based semantic segmentation method addresses the limits of LAMAP and naive Bayes.

### 2.5.1 Addressing Limitations through Neural Network Solutions

Neural networks offer solutions to some concrete limitations of both the naive Bayes and LAMAP method. Due to their relative simplicity, both models are incapable of modelling dependent variables to account for complex geographical phenomena. CNNs elegantly model complex spatial patterns and have been successful at capturing a richer structure of the spatial structure in a local environment compared to local statistical distributions of environmental values [21]. No

explicit parameterization is required because it is learned through a layered architecture to reduce the risk of oversimplification. Instead of considering each site separately, a neural network can integrate data from multiple sites simultaneously to enable collective pattern learning and interaction. This approach reduces the bias introduced when sites are considered separately and their distance weighted. It ensures that the broader context and interactions between various site types are considered, leading to more unbiased predictions. By using techniques such as data augmentation [58], syntactic data generation [51] and semi-supervised learning [74] that elegantly integrate with neural networks, pseudo non-site data can be generated. This reduces the bias introduced by not using non-site data and does not require the extra cost and energy of going into the field. Instead of the many subjective decisions about parameters that must be made in LAMAP, neural networks offer a less biased approach. Through hyper-parameter optimization and neural architecture search the most efficient configurations are found automatically. Major advances in both specialized hardware and software accelerate the learning process of neural networks. Along the hardware side there are Graphics Processing Units (GPUs) and Tensor Processing Units (TPUs) which optimize the necessary computations. Using such hardware does not come with the need for additional specific knowledge as most modern software facilitates easy integration. The modern GPU is not only a powerful graphics engine but also a highly parallel programmable processor featuring peak arithmetic and memory bandwidth that substantially outpaces its CPU counterpart [62]. Besides hardware, frameworks, which offer specialized software, are freely available to automatically optimize the learning process and facilitate easy convenient implementation and operationalization. Examples of such frameworks are Tensorflow [1] developed by google, Pytorch [64] developed by META AI and Keras [17] which was developed independently. They all provide tools for automatic differentiation, memory optimization and model compression.

### 2.5.2 Basics of ANN's

Neural networks originate from the perceptron proposed in 1957 by Frank Rosenblatt. This perceptron consists of a single layer with learnable weights and thresholds between an input and output layer inspired by the learning processes in the human brain [31, 55]. Scientific research, with ups and downs, has made huge strides since then so that ANN's [57] are now the foundation for solutions in multiple disciplines such as interpretation of visual scenes [47, 18] and speech recognition [46]. ANN's flexible adaptive artificial systems and can be applied in approximating real-valued, discrete-valued and vector valued target functions. These systems are able to use data to adapt their internal structure according to an objective function to be learned [28]. Their internal structure consists of essential building blocks: Neurons, layers and activation functions with feed-forward and back-propagation governing the learning process.

#### Neurons and Layers

A neuron is the smallest component of a neural network. This unit converts a real-valued vector (input) according to a linear combination of its values and the neurons weights to a real-valued intermediate result. An activation function which imposes a threshold is used on this result to obtain its output. The calculation for output  $y = o(x_1, \dots, x_n)$  given inputs  $x_1$  to  $x_n$  is given as:

$$\begin{aligned} o(x_1, \dots, x_n) &= f(b + w_1x_1 + w_2x_2 + \dots + w_nx_n) \\ &= f\left(B + \sum_{i=1}^n w_ix_i\right) \end{aligned} \tag{2.6}$$

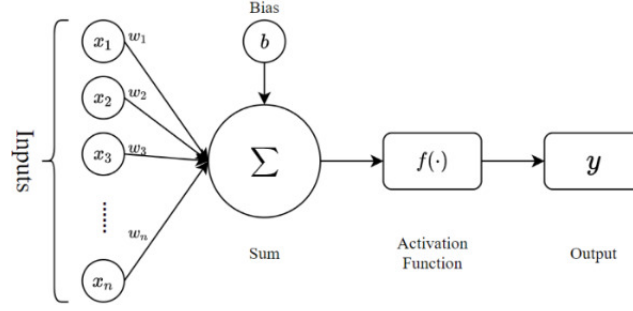


Figure 2.8: Internal structure of an artificial neuron [86]

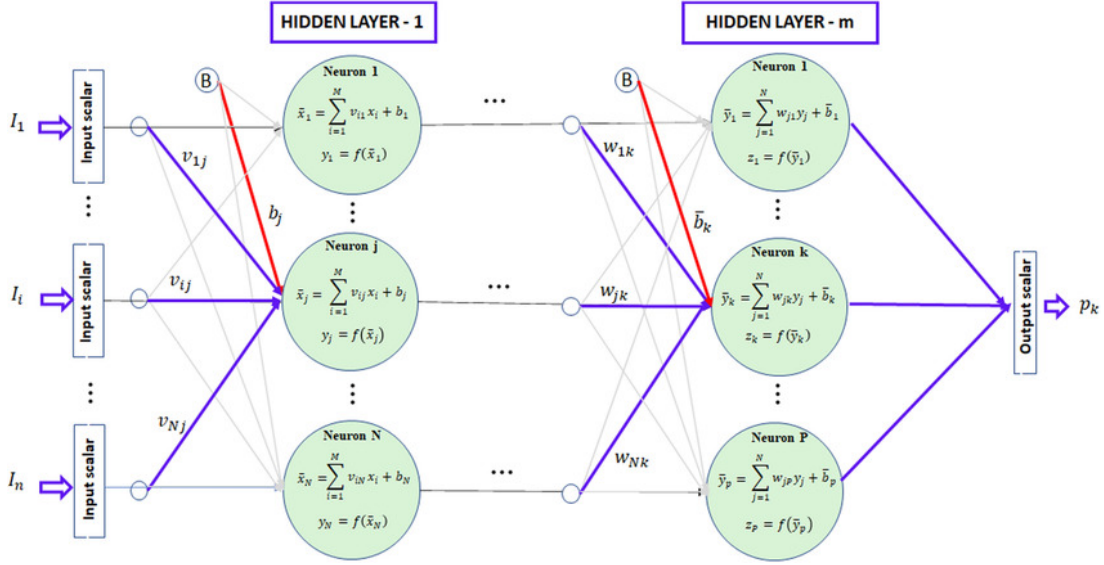


Figure 2.9: Fully connected neural network Architecture [66]

Where each  $w_i$  is a real-valued learnable constant, or weight,  $f$  is an activation function and  $B$  it's bias. Intuitively, the weights determine the contribution of each input parameter to the output. The activation function provides non-linearity and are discussed in the next section. Single neurons can only be used to obtain linear decision surfaces, which severely limits their application. To overcome this limitation, multiple neurons are stacked into a layer. Outputs of each neuron in the layer can be combined through a dedicated activation function to achieve non-linear boundaries. To allow for more abstraction these layers can be stacked, the result is a multi-layer perceptron (MLP).

### Activation functions

As cited above, activation functions play a major role in the ability of an MLP to learn non-linear patterns and model complex relationships in the data. Multiple layers are stacked and passed through an activation function so each layer is capturing a different level of abstraction. Without this type of feature, even a MLP will exhibit the same linear behaviour as a single layer perceptron [26]. Important here is to underscore the differences between activation functions in the hidden



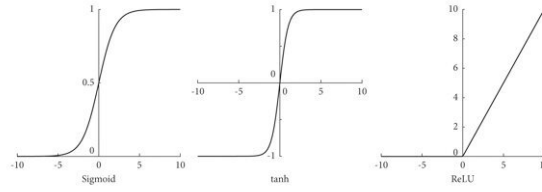


Figure 2.10: Curves of the Sigmoid, Tanh, and ReLU activation functions [36]

layers and output layer. In hidden layers, these functions will introduce non-linearity while in the output layer they facilitate different types of predictions. A range of different functions have been proposed and explored, each with unique properties applicable in different contexts. Here is a brief explanation of some of the most well-known: Sigmoid, Tanh and ReLU visualized in Figure 2.10.

- **Sigmoid:** The sigmoid function is one of the most well-known functions defined as  $\sigma(z) = \frac{1}{1+e^{-z}}$  which effectively squeezed its input between 0 and 1. This naturally represents a probability which is desirable behaviour for instance for an output layer in a binary classification problem where the probability that a data point belongs to a certain class is predicted. However, the use of this function in hidden layers is limited by the saturation problem. Very small or very large numbers end up in the tails of the function so the gradient will approach 0 which will lead to a very slow learning process. This is called the vanishing gradient problem. Additionally, the learning steps might be inefficient as the function is not centred around 0.
- **Tanh:** The tanh function defined as  $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$  centres, unlike sigmoid, around 0 by using a range of -1 to 1 effectively making learning steps more efficient by allowing both negative and positive gradients. Thanks to this property, tanh is more attractive for use in hidden layers but still suffers from the vanishing gradient problem.
- **ReLU:** The Rectified Linear Unit (ReLU) defined as  $ReLU(z) = \max(0, z)$  is widely used in modern neural network applications due to its attractive proportions and improvements over sigmoid and tanh. ReLU maps inputs between 0 and infinity while being easily differentiable which allows for computational efficiency. A negative side effect of this the lower limit on 0 is that neurons can become inactive, output a zero signal for each input and never recover to an active state during training.

Activation functions are still the subject of much study and development. There is no one function fits all problems and the choice of the ‘right’ function is very problem-dependent. In practice, for hidden layers, ReLU is often used due to its simplicity and computational efficiency. For the output layer, a Sigmoid function is often used in binary classification because it converts the network output into a value interpretable as a probability.

### Feedforward and Backpropagation

The training process consists of two steps. During feed-forward, inputs are simply passed through the network. Each neuron transforms its input to output by multiplication with its weights, addition of the bias and application of the activation function. All layers are sequentially passed through until the output layer which gives the final output. Remember here that the network

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**Algorithm 2.5.0** The general gradient descent algorithm; different choices of the learning rate  $\gamma$  and the estimation technique for  $\nabla\mathcal{L}(w)$  may lead to different implementations [71].

---

**Input:** initial weights  $w^{(0)}$ , number of iterations  $T$

**Output:** final weights  $w^{(T)}$

1. **for**  $t = 0$  **to**  $T - 1$
  2.     estimate  $\nabla\mathcal{L}(w^{(t)})$
  3.     compute  $\Delta w^{(t)} = -\nabla\mathcal{L}(w^{(t)})$
  4.     select learning rate  $\gamma$
  5.      $w^{(t+1)} := w^{(t)} + \gamma\Delta w^{(t)}$
  6. **return**  $w^{(T)}$
- 

is only traversed. This output depends on the weights used and the same input gives the same output. The second step called back-propagation is responsible for adjusting the weights and biases with the aim of minimising prediction errors. Using the predicted output after the feed-forward step, the true label and a loss function are used to calculate the difference between prediction and reality. This difference depends on the weights and biases in the network. During back-propagation loss function gradients are computed with respect to the parameters of each neuron using the chain rule. Gradients represent the size and direction of error minimising change proposals and allow for step wise pushing of neuron parameters in the right direction. The overall goal is to update the network so the error is minimised. An optimisation algorithm is responsible of effectively updating the neurons parameters according to its gradient. Examples of such algorithms are *Gradient Descent* and Adam [44]. This 2-step process is iterated for a limited number of steps or until a certain accuracy threshold is reached.

### Overfitting and Regularization

It is said that a models hypothesis overfits its training data when there is another possible hypothesis with a worse fit to the training data but better performance across the entire distributions of examples [57]. Overfitting is a common problem in machine learning where the classic goal is to minimise the error rate on the training data. To counter this, many strategies have been proposed which are explicitly designed to reduce error on test data with the possible effect of increasing error on training data. Collectively, these techniques are known as regularization [26].

### 2.5.3 Introduction to Convolutional Neural Networks (CNNs)

CNN's [47] are a type of ANN specialising in grid-like data such as time-series or images form the basis for many successful practical implementations. Since the data in this dissertation is from Satellite imagery, this will be the focus for subsequent sections. The name convolutional neural network indicates the use of a mathematical operation in the network called convolution. An MLP is called a CNN when the convolution operator is applied in at least one layer instead of general matrix multiplication [26].

#### The Convolution Operation

The convolutional operation [61] can be seen as a specialised neuron. As discussed before, a neuron takes a number of inputs and computes an output using a one-time matrix multiplication. A single convolution takes a full multidimensional array and outputs a feature map computed by

shifting a filter, or kernel, over the input. At each position taken by the filter, an element-wise multiplication takes place between input pixel values in the corresponding filter weights after which they are summed to obtain a single output value. An example of this process is visualized in Figure 2.11. Convolutions improve machine learning networks using 3 main ideas: sparse interactions, parameter sharing and equivariant representations [26].

- **Sparse interactions:** Classical neural network architectures typically have a dense architecture. Specifically, each output neuron uses each input neuron. Matrix multiplication needs  $m \times n$  parameters for  $m$  inputs and  $n$  outputs and their learning algorithms have a runtime of  $O(m \times n)$  per data point. For large input data such as images this is a problem and causes a slow computationally inefficient learning process. CNN's, on the other hand, introduce sparse connectivity by using a kernel smaller than the input. Input images can contain thousands or millions of pixels but interesting information such as object edges can be detected with a small kernel of say 100 pixels. This facilitates a dramatic gain in parameter interference memory and training speed.
- **Parameter sharing:** Each convolution in a convolutional layer uses the same filter parameters in different regions when sliding over the input image. This concept is called parameter sharing. This allows the network to detect the same features in different places in the input. This contrasts a neuron in a densely connected layer where each weight is uniquely applied to a single input neuron. Forward propagation complexity stays the same, but parameter storage space is reduced. Another beneficial property which of parameter sharing for image input is that learned features translation is invariant. Features can be detected regardless of their position within the image
- **Equivariant representations:** A beneficial consequence of parameter sharing is equivariance to translation. This means that for some transformation on the input, such as translation, has a consistent and predictable impact on the transformation of the outcome. In CNNs with images as input, this means that the displacement of an object in the image causes the same displacement in the feature map. The convolutional operation detects the same features regardless of their location so that changes in the input cause the same changes in the output. An important note here, is that this does not inherently apply to all transformations. Examples include colouring and rotations.

### Pooling layers

Pooling layers are used to reduce the input dimensions throughout the network and thus reduce the number of parameters and complexity of the model. This layer is similar to a convolution operation with the difference that its mapping is fixed and thus not learnable. One of the most commonly used are max-pooling layers with  $2 \times 2$  kernels using a 2-step stride of which an example is given in Figure 2.12. Depth size of the input is kept but in this way the input map is scaled down by 25%. This operation is destructive because information is lost. Kernel sizes should therefore be chosen deliberately and are often not set above three. Several alternatives have been proposed such as overlap pooling and general pooling. The right choice is problem dependent.

### Training CNNs

Learning in CNNs is facilitated by the same principles as in ANNs. Both are composed of neurons which self-optimize. Neurons perform an operation on its inputs which in ANNs is a

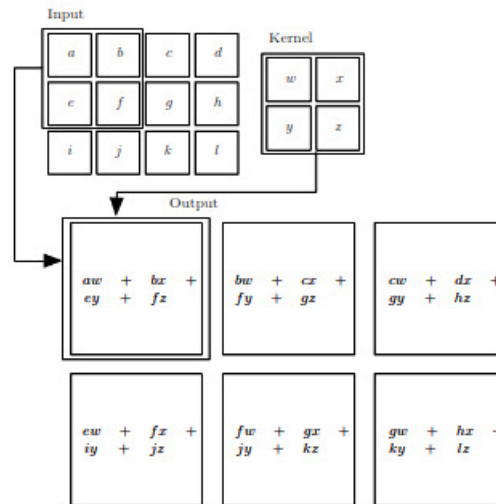


Figure 2.11: An example of 2-D convolution without kernel flipping. We restrict the output to only positions where the kernel lies entirely within the image, called “valid” convolution in some contexts. We draw boxes with arrows to indicate how the upper-left element of the output tensor is formed by applying the kernel to the corresponding upper-left region of the input tensor [26]

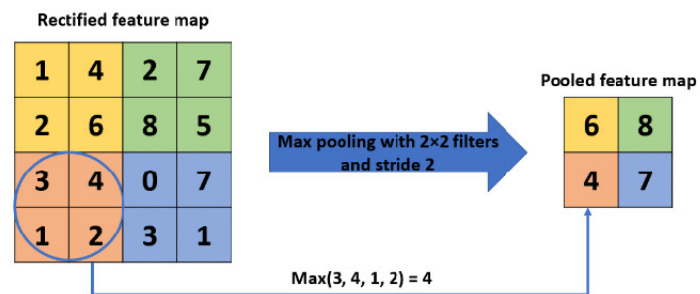


Figure 2.12: Example of Max-Pooling operation [25]

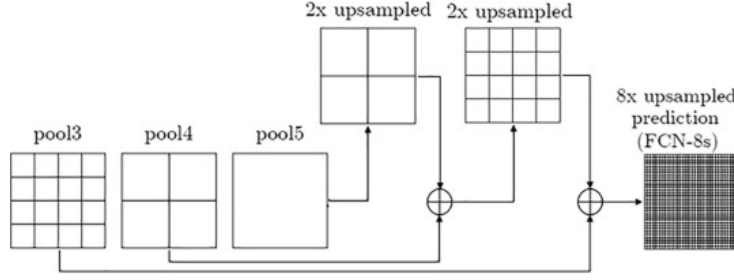


Figure 2.13: The fusion process of FCN to recover the fine details of the segmentation results. [85]

scalar multiplication followed by an activation function while in CNNs this is a convolutional operation followed by an activation function. In CNNs too, the last pair will be responsible for calculating a loss function and all the previous ANN principles are applicable here too. Classically, CNNs consist of convolutional layers, pooling layers and fully connected (FC) layers. The FC layers are the same as discussed in previous ANN section. All these layers are stacked and a CNN is formed.

#### 2.5.4 From Classification to Segmentation

The concepts discussed above in CNNs were applied to a classification problem. However, many real-world problems require pixel level classification, semantic segmentation. For this, the same principles are used with some architecture changes to deal with the increased complexity and granularity of the task. The following sections discuss some of the most common architectures.

##### Fully Convolutional Network

Fully convolutional networks (FCNs) [50] use a classification network as their base but replace the fully connected layers with convolutional layers which produce spatial outputs. Instead of one prediction for its input, a prediction is made for each pixel. We refer to this as semantic segmentation. FCNs consist of 3 steps: multi-layer convolution, deconvolution, and fusion. A  $1 \times 1$  convolution provides the pixel classification but because of the pooling layers, the output will be smaller than the input. Through deconvolution, the original dimensions are again achieved through bi-linear up-sampling. This operation can be seen as a reverse convolution which enlarges its input instead of reducing it. During this up-sampling, detail and information are lost and skip connections are introduced to counteract this loss. Upsampled layer outputs are fused with shallower layers by element-wise addition. The fusion process is displayed in Figure 2.13. In this way the network is able to produce outputs of corresponding dimensions with pixel level predictions using images of varying dimensions as input.

#### 2.5.5 Remote Sensing Segmentation

##### Loss functions

Many deep practical applications of semantic segmentation on remote sensing data use cross entropy as a loss function [85]. The cross entropy for a binary class prediction is represented as follows.

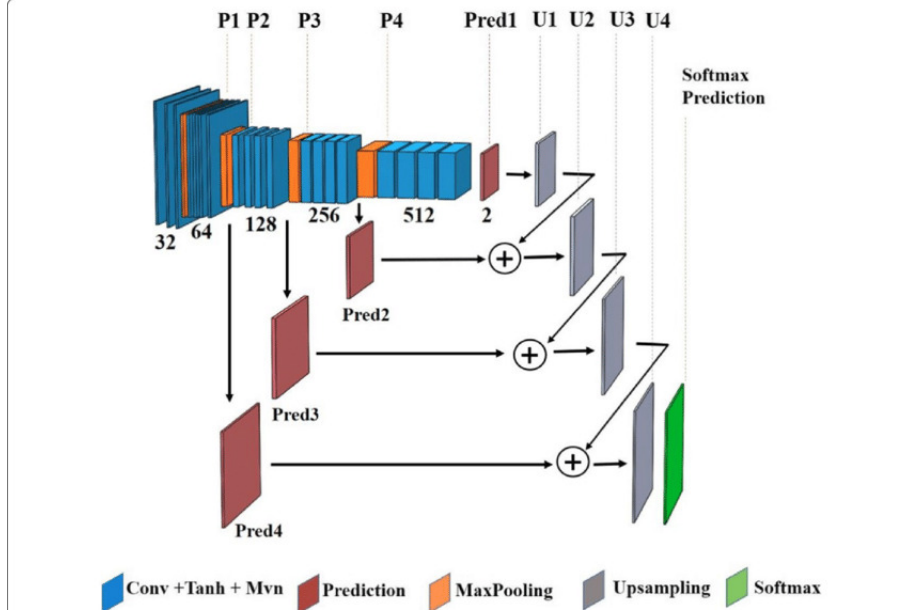


Figure 2.14: Fully convolutional network (FCN) architecture [41]

$$\mathcal{L}_{binary} = -y \log(p) - (1 - y) \log(1 - p)$$

Here  $y$  is the ground truth and  $p$  is the predicted probability of the instance belonging to the target class. When the grounding truth is positive, the output is affected by how close the predicted probability is to 1. When the grounding truth is negative, the density of the prediction to 0 influences the output. For multi-class problems, cross-entropy is defined as:

$$\mathcal{L}_{multi-class} = - \sum_{c=1}^C y_c \log(p_c)$$

Here  $c$  returns the class label. In practice, a softmax function is used for output opportunities  $p_c$ . In problems with unbalanced datasets, the cross-entropy function is modified with the introduction of a weighting mechanism. In a binary context represented as:

$$\mathcal{L}_{weighted} = -y w_1 \log(p) - (1 - y) w_0 \log(1 - p),$$

where  $w$  are weights to balance the classes. The literature identifies three major challenges for the development of deep learning methods for semantic segmentation of remote sensing imagery [85].

1. **Demand for pixel-level accuracy:** A special property of remote sensed imagery in contrast to standard scenic or portrait images is that semantic information is present in each pixel. Not only conventional objects such as buildings and bridges appear in this data but also semantically significant backgrounds such as water masses, roads or fields. Pixel-level accuracy is needed to extract geometric properties such as width and perimeter.

2. **Analysis of non-conventional data:** Many dataset derived from remote sensing sensor are unconventional which provides unique challenges. Point clouds, for example, represent 3D spatial information but have an odd distribution and do not possess the orderly structure required for standard convolutional operations. Another common data source is hyper-spectral images (HSI). These possess 12 to hundreds of channels which thus provide rich spectral details. The richness of this data makes direct application of traditional CNNs difficult for semantic segmentation. Addressing these challenges requires innovative approaches to effectively process and analyze the intricate data provided by these advanced remote sensing modalities.
3. **Lack of training examples:** Training a robust deep neural network model needs a vast number of labeled examples, which are time-consuming and labor-intensive to produce. Data scarcity is a common problem and real-world deep learning application. An additional difficulty in remote sensing applications is the need for a professional's expertise to label different objects. Unlike typical scenic images, crowd-sourcing strategies cannot be used.

### 2.5.6 Semantic Segmentation Applications in Archaeology

Semantic segmentation is used specifically in archaeology in several contexts. Examples using simple object images are corrosion detection in artifacts [70], cultural heritage protection [83] and feature extraction from ancient maps [53]. Using remote sensed data modern semantic segmentation models are able to detect physical archaeological structures [35]. The biggest difference in current corresponding literature are different data sources such as airborne laser scans [5], airborne lidar scans [12] and optical UAV imagery [4]. All of these applications focus on detecting existing archaeological structures such as quarries, temples and stone pits. This dissertation investigates the creation of a predictive archaeological model to predict archaeological potential. This potential represents the probability that a given site contains archaeologically observable traces of civilization. This contrasts with the aforementioned applications where visual identification of architectures is central. As discussed in its dedicated section, computational archaeological predictive models mainly use classical forms of machine learning. Subsequent sections propose an architecture to employ semantic segmentation for predictive modelling using remote sensing data.

## Chapter 3

# Methodology

### 3.1 Data

#### 3.1.1 Sagalassos Archaeological Research Project

Existing and new methods developed in this dissertation are evaluated and compared using the Sagalassos study region. The Sagalassos Archaeological Research Project coordinated by the KU Leuven is subject of study for over a generation. This project covers the archaeological site of Sagalassos and an area of about 1200 km<sup>2</sup>, which roughly matches the ancient city's administrative territory (chora). Located in southwestern Anatolia's Lake District, this area showcases diverse ecological and geographical landscapes. This region is characterized by its fragmented and varied landscape, including fertile plains, narrow river valleys, badlands, hills, plateaus, and steep mountains. Elevation in the study area varies by about 2000 meters, contributing to a diverse range of vegetation from semi-arid steppes to deciduous woodlands and Mediterranean-like zones at higher elevations [65]. In this thesis, individual predictive models are built for 7 different time periods during which occupation in the study area has been demonstrated: Late Prehistory (6500-2500 BCE), Iron Age-Archaic (1150-546 BCE), Achaemenid-Hellenistic (546-25 BCE), Roman Imperial (25 BCE-300 CE), Late Antique (300-700 CE), Byzantine (700-1200 AD), and Late Ottoman (1700-1921 CE). Among them, occupations from different time periods are distinguished by demonstrable differences in settlement patterns and material culture [81]. For each of these time periods, a selection of independent variables is made which experts believe to have archaeological potential influence. The selection is shown in Table 3.1 and are the same as in the application of LAMAP [81] in the study region to facilitate an introductory comparison.

#### 3.1.2 Sagalassos Site Database

25 years of archaeological research in the Sagalassos region is summarised in the 'Sagalassos Site Database'. The 2018 version of this database [81] describes 264 unique sites over an area of about 1450 km<sup>2</sup> on which a total of 463 sites belonging to different time periods are indicated. Some features important for this dissertation found in the database are geospatial coordinates, relative dating and site type. Sites included in the database are defined as 'places where archaeological activity has been observed'. Not only sites characterised by prominent structures such as stone remains of historical buildings, but rather sites containing various artefacts such as fragments of ceramic vases or stone tools dated by experts have been included. For each site, the amount of information strongly depends on the method of investigation. Excavations were only carried



|                          | LP | IAA | AH | RI | LA | B | LO |
|--------------------------|----|-----|----|----|----|---|----|
| Elevation                | x  | x   | x  | x  | x  | x | x  |
| Aspect                   | x  | x   | x  | x  | x  | x | x  |
| Slope                    | x  | x   | x  | x  | x  | x | x  |
| Convexity                | x  | x   | x  | x  | x  | x | x  |
| Distance to Drainage     | x  | x   | x  | x  | x  | x | x  |
| Distance to AH Roads     |    |     | x  | x  | x  | x |    |
| Distance to RI Roads     |    |     |    | x  | x  | x |    |
| Distance to City Centers |    |     | x  | x  | x  | x |    |

Table 3.1: Table shown the used predictive features per time period: LP (Late Prehistory), IAA (Iron Age-Archaic), AH (Achaemenid-Hellenistic), RI (Roman Imperial), LA (Late Antique), B (Byzantine) and LO (Late Ottoman)

out at a very small proportion of sites while the majority were subjected to surface prospection which can be divided into extensive and intensive methods. The number of known positive sites varies for each time period: Late Prehistory 48, Iron Age-Archaic 29, Achaemenid-Hellenistic 67, Roman Imperial 89, Late Antique 114, Byzantine 52, Late Ottoman 64.

### Extensive surveys

A survey is extensive when it does not involve a systematic approach. Archaeologists visit and investigate sites of potential value on foot using historical maps or verbally obtained information from local residents. 135 sites (46.7%) of the sites in the Sagalassos Site Database are the outcome of an extensive approach. An obvious drawback lies in the bias of this method. Historical maps and oral information often point to highly visible sites with monumental features from the same time period, making them over-represented.

### Intensive surveys

Unlike Extensive surveys, a systematic method is used here. This type of survey aims to document a wide type of landscapes without using prior knowledge about discovered sites and can take different forms. In grid surveys, the survey region is divided into squares of equal size. Each square is then the subject of a systematic survey such as the free collection of artefacts over a period of time or multiple helpers work together to collect all artefacts within this square. Another method and source of data are transected surveys. This involves defining a parallel running line at equal intervals, which is walked by a researcher who collects artefacts within an agreed distance from the running line. The artefacts consist mainly of lithics such as fragments of ceramic vases or stone tools. The collected material is dated by experts so that that site can be labelled with the corresponding time period.

### 3.1.3 Geospatial Data

Geography has long been a guiding force in human history and continues to shape our future. From the stone-age hunters who anticipated the location of their quarry to early explorers who relied on their understanding of geography for survival, and even to modern societies where daily activities are influenced by spatial considerations, geography remains paramount. To integrate geography into the technological realm, GIS [10] have emerged. These systems empower us to collect and manipulate spatial data, providing invaluable insights into various phenomena. At the

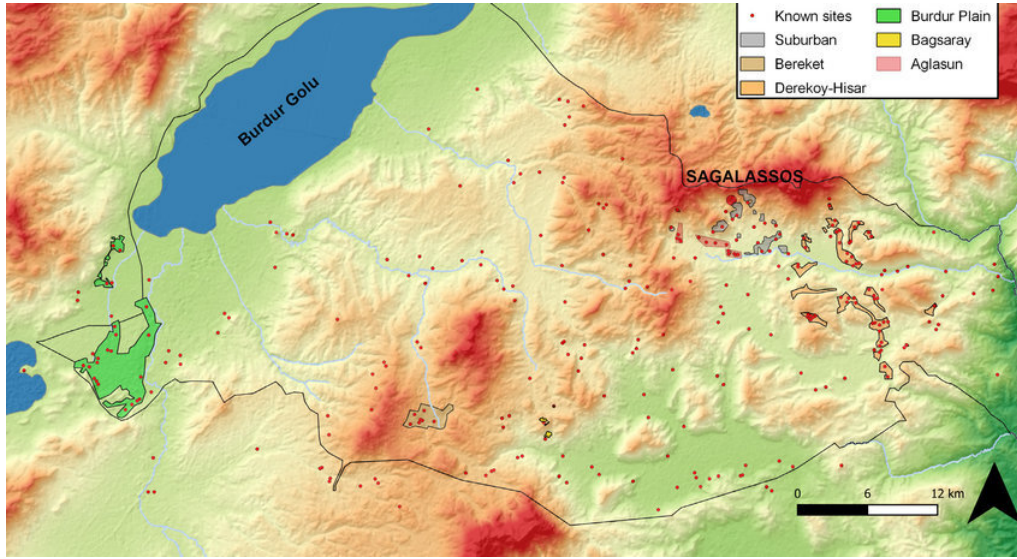


Figure 3.1: The Sagalassos study region with the locations of the various intensive survey areas and all known archaeological site locations indicated. The extensive survey was carried out within the entire study area [82].

heart of GIS lie two fundamental concepts: "what" and "where." The "what" encompasses features such as elevation or population size, while the corresponding "where" is indicated through Coordinate Reference Systems (CRS), which utilize coordinates to pinpoint the location of these features on Earth's surface. A valuable feature of GIS is that new data can be easily extracted. Often this new data is more valuable according to the context and no new measurements are needed. A good example and application of this in this dissertation is deriving new features from a digital elevation model (DEM) which provides according to a certain resolution for each pixel its elevation. This model can be used to derive features such as steepness (slope), direction of ground facing (aspect) or drainage routes. Landscape variables are a strong indicator of human occupation of an area. Humans adapt to ecological changes and factors based researched behaviours such as minimising and optimising energy consumption [80]. This dissertation uses parameters for which a correlation with human occupation has been determined in the Sagalassos study area [81]. These are discussed below.

### Digital Elevation Model

The Terra Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) Global Digital Elevation Model (GDEM) Version 3 (ASTGTM) provides an elevation model with global coverage with a spatial resolution of about 30x30 metres [20]. Its development is a collaboration between National Aeronautics Space Administration (NASA) and Japan's Ministry of Economy, Trade, and Industry (METI). The combined satellite imagery was collected between March 1 2000 and November 30 2013. Clouds are masked out and residual bad values and outliers are removed [2]. The ASTGTM forms the basis of the geospatial data used and is represented by a regular square grid. In this grid, elevations (Z values) are available as a matrix of points equally spaced in two orthogonal directions. Other features such as aspect, slope, and profile are calculated based on the z-values, utilizing the magnitudes and patterns of change in Z across

| Variable                    | Description                                      | Importance                                                                                                                                   |
|-----------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Elevation                   | Elevation above sea level                        | Climate (Temperature, vegetation, ..) [54] and defense [63].                                                                                 |
| Slope                       | Rise relative to horizontal distance             | Water flow; flooding, erosion, travel costs, construction suitability, geology, insolation, soil depth.                                      |
| Aspect                      | Downhill direction of steepest slope             | The sun is the earths energy source. More sunlight is beneficial for insolation, temperature, vegetation, soil characteristics and moisture. |
| Profile                     | Curvature parallel to slope direction            | Erosion, water flow acceleration.                                                                                                            |
| Distance to water resources | Euclidean distance to closest probable waterways | One of the most universally accepted environmental variables is the distance to water sources.                                               |
| Distance to roads           | Euclidean distance to closest ancient road       | Erosion, water flow acceleration.                                                                                                            |
| Distance to city centers    | Euclidean distance to closest city center        | These center accommodated larger population and are therefore expected to be build on suitable soil [78].                                    |

Table 3.2: Description of Variables adapted from [10] and [45]

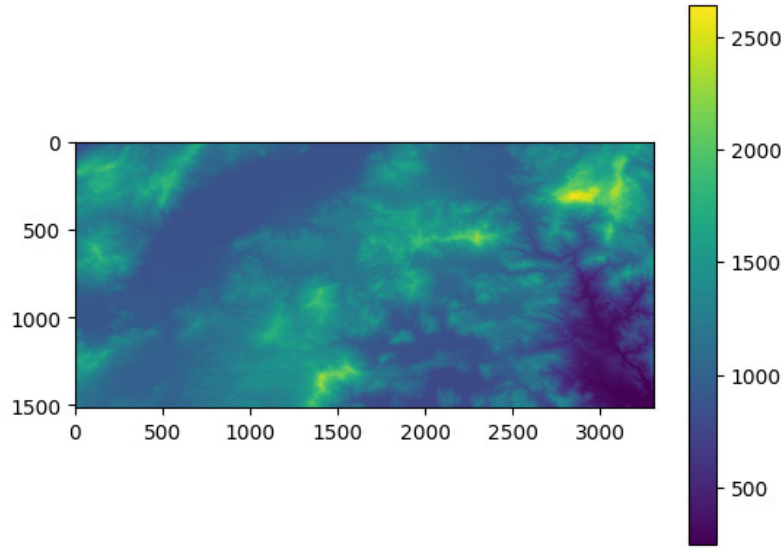


Figure 3.2: Sagalassos study region DEM plot. Color indicates pixel elevation in meters.

the grid. Each terrain attribute can be computed through a mathematical operation applied over a moving window across the DEM. Derivations are created using Richdem, a library that provides digital elevation model hydrological tools using parallel processing and state-of-the-art algorithms [6]. For the project, a subset of 1515 x 3302 pixels is used which covers the entire Sagalassos study region 3.2.

### Slope

Slope is the change in elevation (a rise) according to a change in horizontal position (a run). Slope equals 45 in degrees if the rise equals the run. Conversion of rise/run to degrees can be done by applying the tangent trigonometric function. Grid cells contain data at regular intervals at irregular true distances. Slope is typically measured in the direction of the steepest elevation change as seen in 3.4. Slopes change in a complex way across the landscape and calculations rely on relative changes around each individual cell which makes for complex calculation. In the Richdem library an algorithm is provided based on [32]. Horning et al. compute the slope in a given pixel via a central difference estimation of a surface fitted on the pixel and its neighbours and gives the rise over run. Intuitively, the elevation model can be represented as a three-dimensional plane with x and y the coordinates and z the elevation. For each location, a certain rise is added to the z-axis. In each direction, the run is calculated being the number of pixels that can be stepped until the location again belongs to the surface. For example, for a certain rise if the slope is steep, the run will be low resulting in a high rise/run. Rise over run can be transformed to an easier-to-interpret form such as slope percentage via conversion formula:  $\% = (rise/run) * 100$ .

### Aspect

The aspect of a pixel represents the direction of its steepest downhill slope, typically described by the azimuth angle where 0 represents north, increasing in a clockwise fashion until reaching

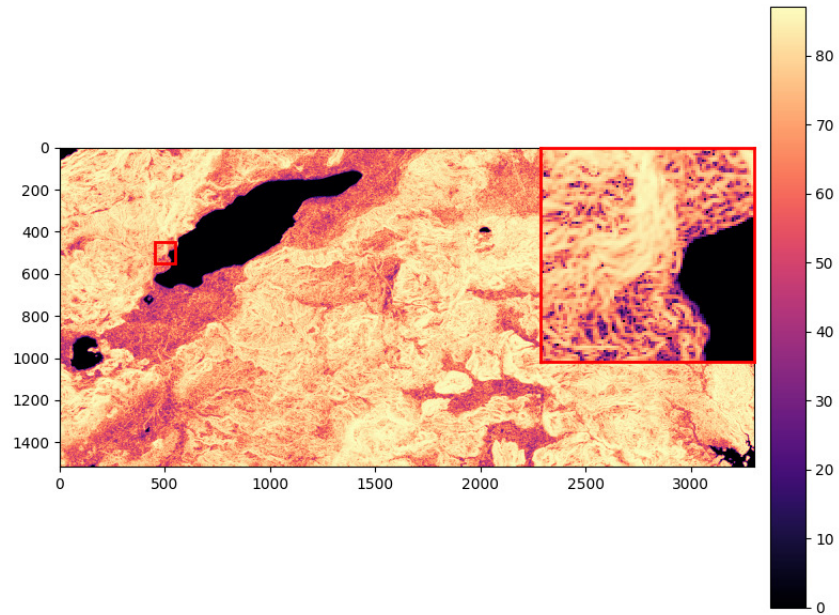


Figure 3.3: Sagalassos study region slope plot. Color indicates pixel rise/run.

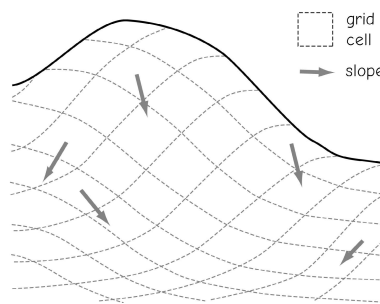


Figure 3.4: Slope direction, shown as grey arrows for some pixels, often change substantially among cells on a raster surface. All surrounding cells are taken in account for calculation [10].

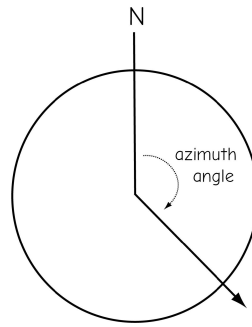


Figure 3.5: Aspect reported as an azimuth angle measured clockwise in degrees from north [10].

360 degrees [32] as seen in Figure 3.5.

### Convexity

Profile curvature of a pixel is another important terrain feature that, like slope and aspect, can be derived from a DEM. Profile curvature is an index indicating the shape of the slope and the steepest downward slope. This can be represented by imagining a plane intersecting the earth, with the line of steepest descent (aspect) on the plane. The surface traces a path along the face of this plane and the curvature is defined by the shape of this path. A negative curvature (3.7 A) indicates an upward slope, a positive one (3.7 B) indicates a downward slope while zero (3.7 C) represents a linear slope [32, 10]. It should be noted that other curvatures exist. Plan curvature is the profile shape in the local direction at right angles to the steepest slope. This feature is not considered in this dissertation as well as it is not used by P. Willit in [81] but may be applicable in further research.

### Proximity to Drainage

A fundamental part of geoscience and anthropological research into the past are natural drainage networks. Access to running water drives ecological progress in an area and supports a human population in many ways making its presence a strong indicator of human occupation. Conversely, ancient settlements are also a strong indicator of certain environmental conditions such as water availability. Drainage networks are formed by the lines on which gravitational drag combined with slope through fluvial processes transports water. The calculation is based on a 3x3 grid centred around each pixel of interest which allows the method to scale easily to large matrices such as the entire study region [60]. From an anthropological point of view, it is interesting for ancient civilisations to develop close to these natural flows. In the next step, the drainage network is transformed into a model in which the euclidean distance to the closest flow is calculated for each pixel.

### Proximity to City Centers and Ancient Road Networks

Apart from environmental features, cultural variables are also used for which differ period. These variables are distance to urban centres for periods known for large-scale settlements (Achaemenid-Hellenistic, Roman Imperial, Late Antique, and Byzantine) and also distance to ancient roads and the least-cost paths connecting them.

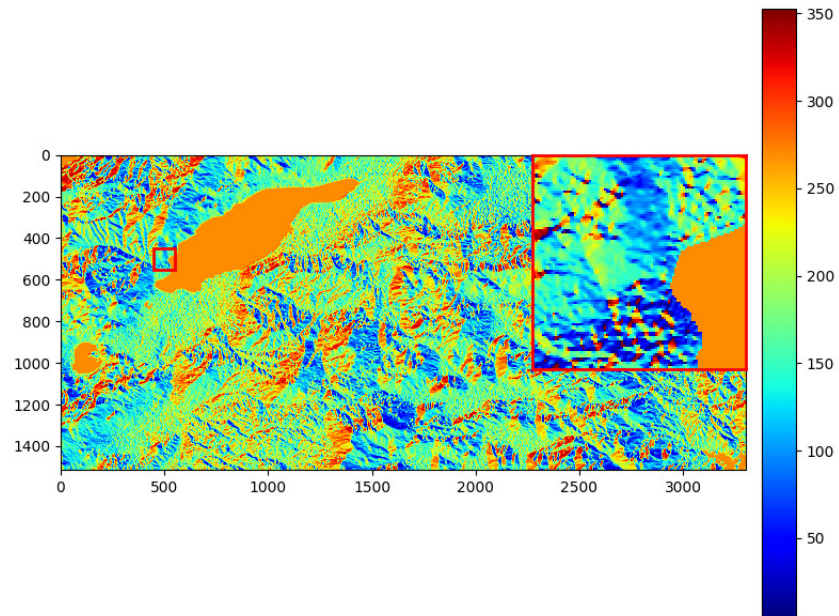


Figure 3.6: Sagalassos study region aspect plot. Color indicates slope orientation in degrees.

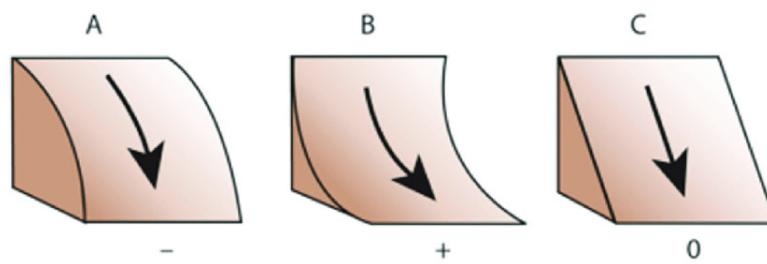


Figure 3.7: The principle of calculating profile curvature [22].



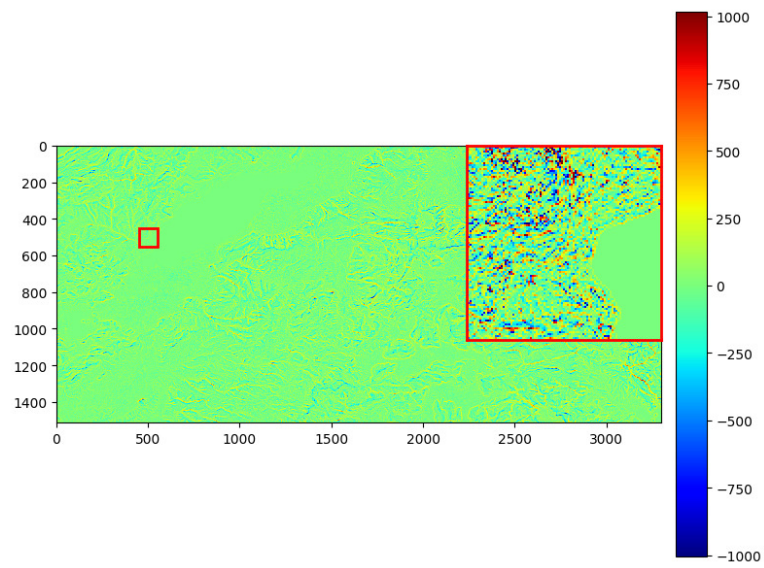


Figure 3.8: Sagalassos study region curvature plot. Color indicates direction of slope curvature

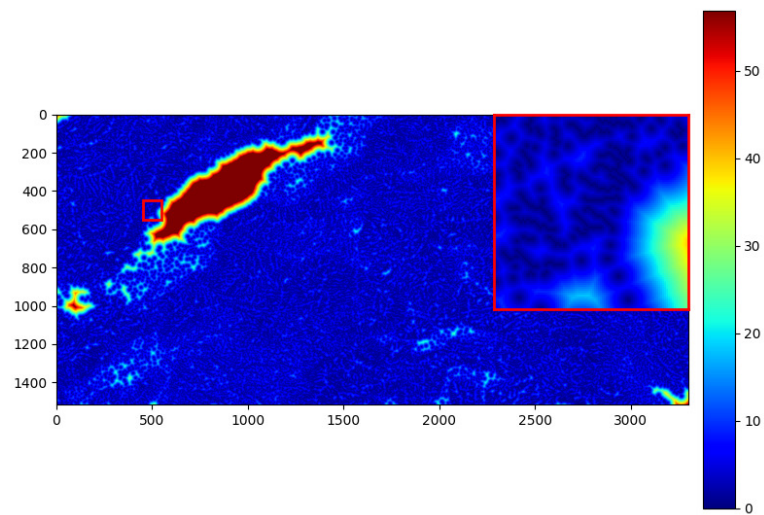


Figure 3.9: Sagalassos study region distance to drainage plot. Color indicates euclidean distance to the closest flow within the raster.



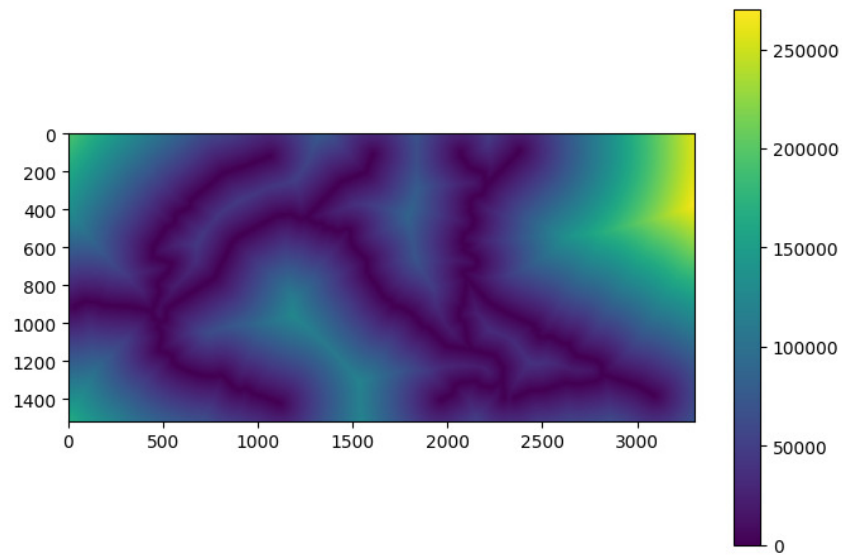


Figure 3.10: Sagalassos study region Roman road networks. Map shows the euclidean distance from every pixel to the closest Roman road.

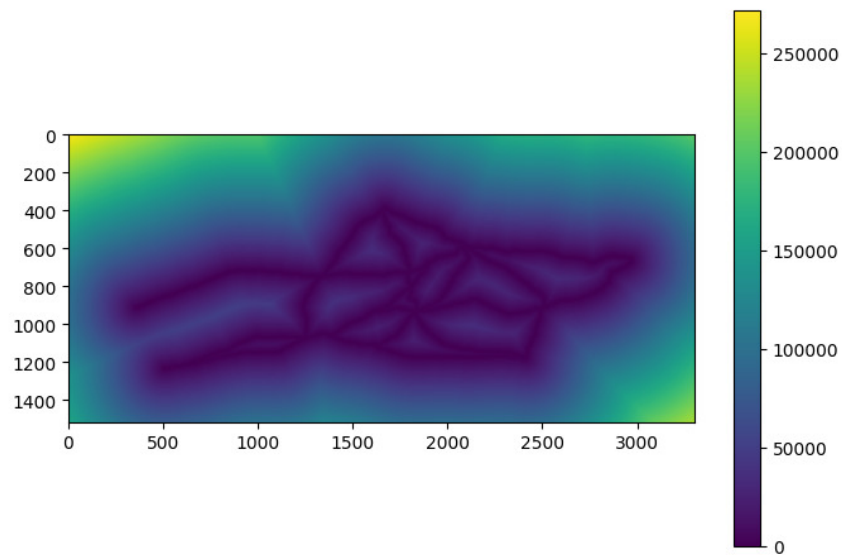


Figure 3.11: Sagalassos study region Hellenistic road networks. Map shows the euclidean distance from every pixel to the closest Hellenistic road.

| Layer | Type               | Input Depth | Output Depth | Kernel Size          | Output Size |
|-------|--------------------|-------------|--------------|----------------------|-------------|
| 1     | Conv2D + LeakyReLU | 8           | 32           | 3x3, stride 1, pad 1 | 16x16       |
| 2     | MaxPool2D          | -           | -            | 2x2, stride 2        | 8x8         |
| 3     | Conv2D + LeakyReLU | 32          | 32           | 3x3, stride 1, pad 1 | 8x8         |
| 4     | Conv2D + LeakyReLU | 32          | 32           | 3x3, stride 1, pad 1 | 8x8         |
| 5     | Conv2D + LeakyReLU | 32          | 32           | 3x3, stride 1, pad 1 | 8x8         |
| 6     | Conv2D + LeakyReLU | 32          | 32           | 3x3, stride 1, pad 1 | 8x8         |
| 7     | ConvTranspose2D    | 32          | 32           | 2x2, stride 2        | 16x16       |
| 8     | Conv2D             | 32          | 3            | 1x1, stride 1, pad 0 | 16x16       |

Table 3.3: Overview of the FCN Architecture

## 3.2 Adapting Semantic Segmentation to the Research Area

This section explains the semantic segmentation model with the aim of predicting potential archaeological sites used in this research. For building neural networks in any context, deliberate decisions need to be made for components such as hidden layer activations and loss functions. An important feature of the context in this dissertation is the relative scarcity of labels. This important property must be taken into account when constructing the model.

### 3.2.1 General Architecture

The chosen architecture for semantic segmentation in this dissertation is a simple fully convolutional neural network (FCN) consisting of an encoder and decoder unit and uses Leaky Relu activation functions. Each layer uses a kernel size of 3x3, a stride of 1x1 and a padding of 1. The encoder uses max-pooling to down-sample its input by 2x effectively increasing its receptive field. After this the decoder restores the inputs spatial dimensions by using a transposed convolution. For example, for predictions for the late antique time period, the input consists of a 32x32 image with a depth of 8 for the selected geographical characteristics: elevation, aspect, slope, convexity, distance to waterways, distance to Hellenistic roads, distance to roman roads and distance to known Late Antique centres. During training, the input is accompanied by its 32x32 one-dimensional mask with labelled pixels corresponding to known sites and/or known non-sites from this period in this region. An example of the layered input with its ground truth mask is given in Figure 3.12. The model is trained individually for each time period for 15 epochs each consisting of 256 randomly selected samples in batches of 4 with a learning rate of 1e-3. Only samples containing at least 1 labelled pixel are used as considering the model ignores all unlabelled pixels. Each input layer is re-scaled to [0, 1] so that the learning process is not affected by the differences in units used between, for example, elevation in metres and aspect in degrees. The architecture used is summarised in Table 3.3.

### 3.2.2 Specialised library

To introduce semantic segmentation to the geospatial setting of modelling archaeological potential, a specialised library built on the well-known Pytorch was used. Torchgeo offers a collection of datasets, transformers samplers and models that have been specially developed and optimised for easy integration into geospatial applications. It tries to provide solutions to the complexity of often fragmented data offered in different projections and resolutions. In the context of

this thesis, for example, environmental characteristics extracted from GIS techniques are offered in EPSG:4326 while the Sagalassos dataset indicates archaeological sites in EPSG:32636. EPSG stands for European Petroleum Survey Group. The numbers after this designation form a unique identifier for coordinate systems. EPSG:4326 is a geographical coordinate system which uses latitude and longitude on a spherical model of the earth with degrees as the unit. In contrast, EPSG:32636 is a projected system specific to the universal transverse Mercator zone 36N in which metre is used as the unit. The latter system includes, among others, countries such as Egypt, Finland, Ukraine and also Turkey. Unlike EPSG:4326, EPSG:32636 provides a flat 2-dimensional space which ensures accurate distances in this region.

### 3.2.3 Activation function

Rectified linear units (ReLU) are an excellent choice for hidden units in neural networks [26]. One drawback, however, is that they cannot learn via gradient-based methods on examples for which their activation is 0. However, there are plenty of generalisations of ReLU which guarantee a gradient everywhere in the network. One of them, Leaky ReLU [52] gives a small gradient to negative values. This not only solves the dying ReLU problem but also balances this activation function to speed up the training process [43]. For small datasets, such as the dataset used in this study, it has been shown that using ReLU like functions is even more important than learning the weights in hidden units [59]. This motivates the choice of leaky ReLU as activation functions in hidden units in this dissertation.

### 3.2.4 Loss function

The use of cross entropy loss, which was introduced in section 2.5.5, in the model is motivated by its large shown improvements in performance for models with sigmoid and softmax outputs [57]. These models where the classical mean squared error loss is used suffered from saturation and a slow learning process. The training samples used in this dissertation exhibit binary target values. The class of non-archaeological sites are labelled by 0 while the class consisting of positive archaeological sites are labelled 1. It may be more valuable to model the underlying target function by outputting the probability that a sample belongs to a class rather than a direct class prediction. It can be shown that in situations where these probabilities are estimated, the maximum likelihood probabilities are given by a network which minimises its cross entropy [57].

### Sample of Late Antique input channels and mask

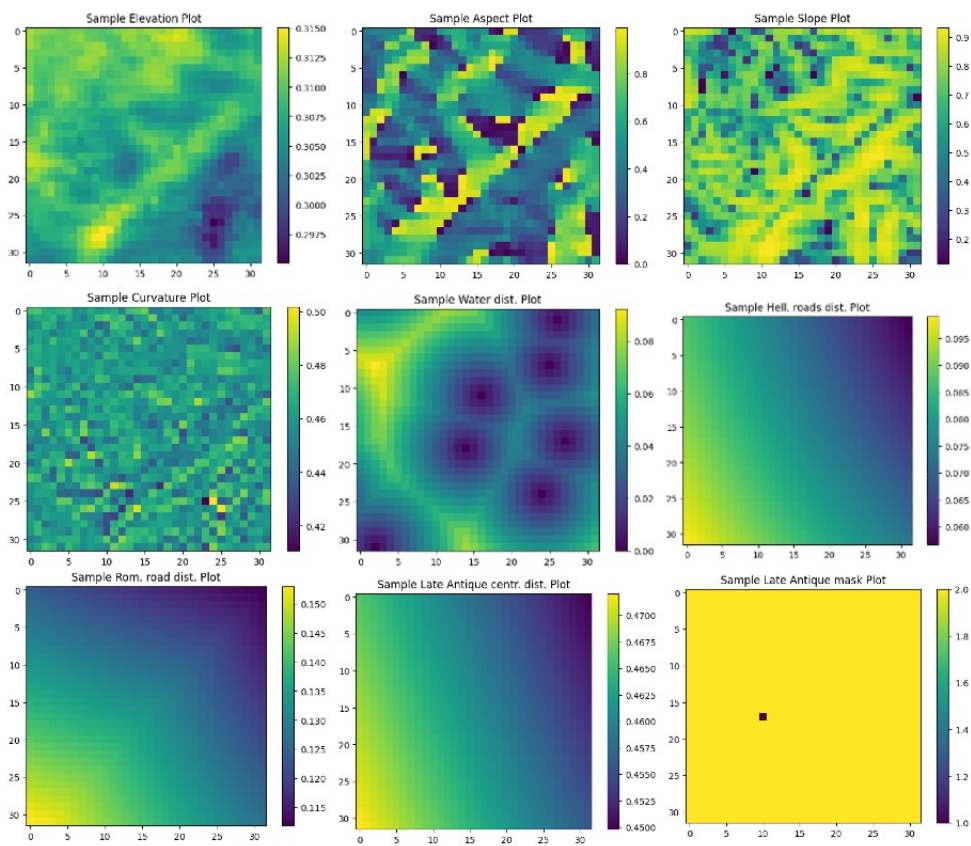


Figure 3.12: Plots showing a semantic segmentation training example.

# Chapter 4

## Results

### 4.1 Overview

In this chapter, the results discuss the capabilities of the different models to model archaeological potential. This can be divided into 3 subsections. Firstly, LAMAP and naive Bayes will be directly compared using the data collected for LAMAPs validation in the Sagalassos survey area [81]. Next, the individual binary classification capacity of naive Bayes is tested. Finally, an overall comparison between the site occurrence predictions of naive Bayes, semantic segmentation and LAMAP follows. The following sections introduce test metrics and frameworks used in the various analyses.

#### 4.1.1 Cross-Validation

With the exception of a direct comparison of naive Bayes and LAMAP in section 4.2, k-fold cross validation [7] is used to gain insight into the predictive capabilities of the different methods. K-fold cross-validation is a widely used technique within machine learning in which the entire dataset is split into k subgroups or ‘folds’ of approximately the same length. The models are trained k times by using a different fold each time as validation and the remaining k-1 folds as training data. At the end of the k iterations, each fold has been used exactly once as validation. Specifically, for our data, this means a split of known site and known non-site into k non-overlapping subgroups. Imagine using a single train/test distribution. The results will be dependent on this distribution because, by chance, the training instances can be easily distinguished from the test instances. Conversely, the training set may also consist of biased training data which poorly represent the problem making it harder to fit a general hypothesis. An advantage of k-fold cross-validation is that it gives a more robust estimate of performance by averaging over different train/test distributions for chosen metrics.

#### 4.1.2 Precision, Recall and F1-score

In the section 4.3, the global capacity for binary site prediction of naive Bayes is tested using 4-fold cross validation with precision, recall and F1-score as metrics [26]. Precision represents the ratio of positive predicted samples which were actually positive according to the ground truth. High precision indicates a model which predicts few false positives. Recall, on the other hand, reflects the model’s ability to include all positive instances. High recall indicates the ability to correctly classify a high portion of the positive classes. The harmonic mean between precision

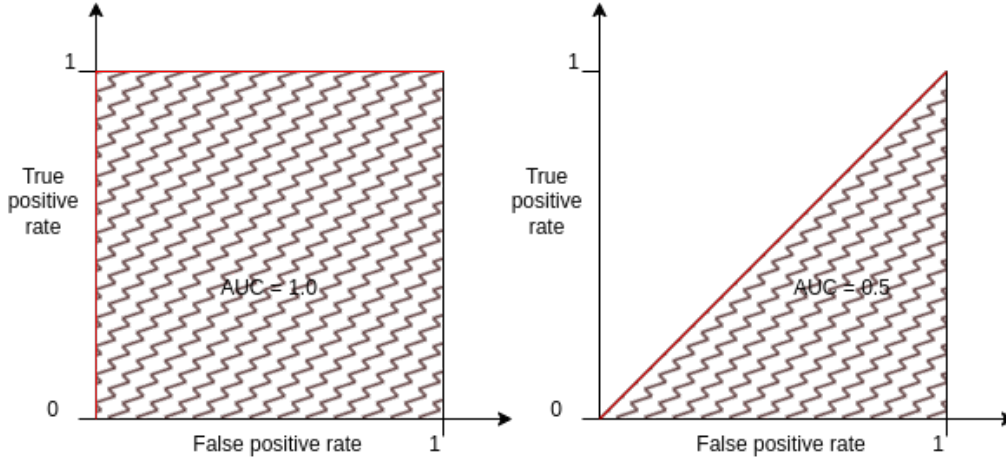


Figure 4.1: ROC curve for a perfect binary classification model with an area under the curve of 1.0 on the left. On the right the ROC curve for a random prediction which will result in an AUC of 0.5.

and recall gives the F1 score. When used in k-fold cross validation, the metrics are calculated separately on each fold and averaged. In this way, attempts are made to understand the strengths and weaknesses of the model across different subsets.

#### 4.1.3 Receiver Operating Characteristic and Area Under the Curve

In section 4.4, naive Bayes, LAMAP and semantic segmentation are compared using receiver operating characteristic (ROC) curves and area under the curve (AUC) metrics collected by evaluating the models on the same 4-fold data. Both are tools in assessing a classification with the favourable feature being their ability to provide insight into different tipping points by which an example is considered positive. ROC curves provide a graphical representation of model performance by plotting the true positive rate (TPR) versus the false positive rate (FPR) over a varying classification threshold. TPR represents the proportion of labelled positive instances that were labelled as positive by the model. Note that this is actually the recall. FPR, on the other hand, represents the proportion of negative data points incorrectly labelled as positive data points by the model. For a binary classifier model, the ROC curve provides a visualisation of the trade-off between TPR and  $1 - \text{FPR}$ . Figure 4.1 shows an example ROC curve for a model with perfect performance. Perfect classification will lead to a TPR of 1 (all positive instances are labelled as positive) and an FPR of 0 (no negatives were labelled as positive). AUC quantifies the entire capacity of the model by calculating the area under the ROC curve. This area has a range of 0 to 1. An AUC of 0.5 indicates a model with no discriminative power. For example, a model which makes a random guess for each of its inputs yields an AUC value of 0.5. This is illustrated in Figure 4.1 on the right. In contrast, visualised in Figure 4.1 on the left, an AUC of 1 indicates error-free classification. When comparing models, preference is given to models with AUC values closer to 1. These models are, according to an AUC comparison, better able to distinguish positive and negative instances

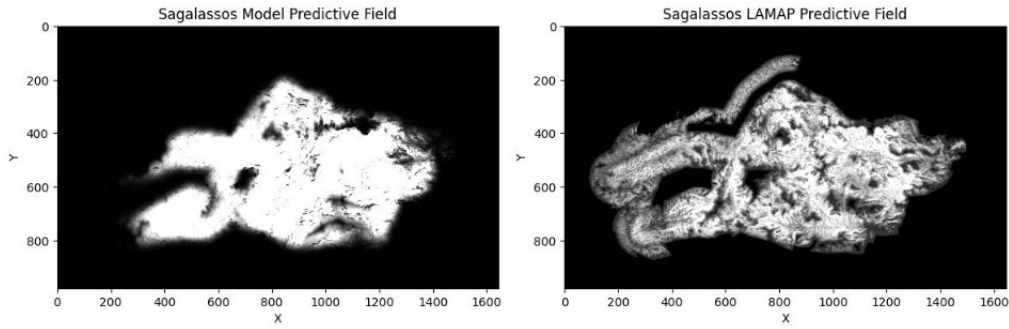


Figure 4.2: The naive Bayes predictive surface for the Late Antique time period (left) next the the predicted LAMAP surface for the same time period (right).

## 4.2 Testing naive Bayes on LAMAPs Survey Data

In Figure 4.2 the predictive surface of LAMAP is shown next to a predictive surface of naive Bayes. In this section LAMAP is compared with the naive Bayes method and a preliminary test for the applicability of naive Bayes is carried out. An important note here is the way the naive Bayes predictions are used in this section. As discussed earlier, in section 2.4, both positive and negative class instances are available for the naive Bayes method. Underlying, however, naive Bayes models the different classes individually. For predictions, against each class, the probability by which the data point belongs to this class is calculated. The class with the highest predicted probability gives the final prediction. In LAMAP, only positive instances are used as the basis for predictions on unseen data. It is assumed that relatively smaller predictions indicate less archaeological potential and larger ones indicate more potential. In the comparison in this section this is also applied to naive Bayes by using only the calculated probability by which a new data point belongs to the class of archaeological sites. This facilitates a comparison of LAMAP and naive Bayes based on the same data consisting of only known archaeological data points. The naive Bayes method is evaluated and compared on the data collected in the reliability analysis of LAMAP in the study area. This analysis is discussed in next section.

### 4.2.1 LAMAPs Reliability Survey

The LAMAP method was extensively tested in the study area using a ground-truth survey [81]. For these surveys, LAMAP's cell predictions were randomly selected, examined and their results compared with its corresponding predictions. The algorithm's output has a range of 0-5 indicating the potential of a pixel. The survey aimed to record multiple cells exciting across the range of potentials for each period. The fieldwork was conducted by three researchers who walked 50 metres with mutual spacing of 25 metres and man-made materials collection within 1 metre of their walking line. Collected objects were counted, described, documented and dated by project ceramology experts Jeroen Poblome (research group Archaeology, KU Leuven), Dries Daems (research group Archaeology, KU Leuven), and Ralf Vandam (Faculty of Arts and Philosophy, Vrije Universiteit Brussel). A total of 101 cells were examined. Based on these data, LAMAP shows strong predictions for the Achaemenid-Hellenistic and Late Prehistoric periods. The increase in predicted LAMAP class for these periods has a high correlation with the probability of finding a site. Weaker correlation was shown for the Iron Age and Byzantine models. One result of this research is a collection of ground-truth cells which can be used to test the reliability of other

models. It should be noted that the cells are chosen considering an equal distribution across LAMAP classes. By equal representation, a fair study is made on the observed versus predicted archaeological potential by LAMAP.

### 4.2.2 LAMAP Versus naive Bayes Results

As discussed in section 2.4.2, naive Bayes offers big gains in computational speeds and resource needs compared to LAMAP. In the following, we extend our comparative analysis by a preliminary comparison between LAMAP and naive Bayes. Naive Bayes predictions are sampled for the selected cells in the LAMAP study. The predictions, which originally range from 0 to 1, are divided into 5 bins to facilitate comparison to LAMAP. An idea used in LAMAP is that cells in the neighbourhood of positive cells help the predictive model and so are included as positive. For this comparison, the naive Bayes model also uses these neighbouring cells. In subsequent subsections, LAMAP and naive Bayes are compared on 2 tasks. First their potential to discover sites and second their prediction correlation with the number of artifacts found. LAMAPs performance and the testing methods in this comparison are taken from the results by P. Willet in [81]. Although the data used was collected according to an equal distribution across LAMAPs prediction classes, the following section will make a first study of the potential of naive Bayes in predicting archaeological potential

#### Probability of Presence

To compare the capacity of the naive Bayes and LAMAP method to predict the presence or non-presence of artefacts, a logistic regression model is constructed using the model's prediction as predictor variable. The aim is to assess the effect of a predictor (the predicted class) on the outcome (the presence or absence of a site). For this analysis, the time periods were considered as a whole. In Figure 4.3 bar plots show the empirical probability of finding something from each time period given a predicted naive Bayes or LAMAP class. In an ideal world, the bar plots would show a linear upward trend for an increasing predicted class. For the naive Bayes predictions, the logistic regression model gives a positive coefficient for the predictor. An increase of 1 in the predicted class gives a 6% increase in the probability of a site. In LAMAP, this is an increase in the probability of 27%.

#### Artifact Count Correlation

Using a Poisson regression model, the correlation between the predicted naive Bayes class and the number of artefacts found is examined. Again, the predicted class serves as the predictor. Figure 4.4 visualises a comparison between the number found for each class for each time periods between LAMAP and naive Bayes. According to the application of the model, again the predicted class is positively correlated with the number of fragments found. An increase in predicted class of 1 increases the number of predicted fragments by 18% whereas 52% in LAMAP. This result shows the desired characteristic in naive Bayes predictions that an increase in predicted class causes an increase in the number of artefacts found.

#### Remarks

Previous analysis shows a preliminary test of the effectiveness of the naive Bayes method in the study area using the data collected in a test designed for the LAMAP method. It is important to re-emphasise that the cells studied were selected to obtain an equal distribution across LAMAP prediction classes. For a fair comparison between LAMAP and naive Bayes, a new



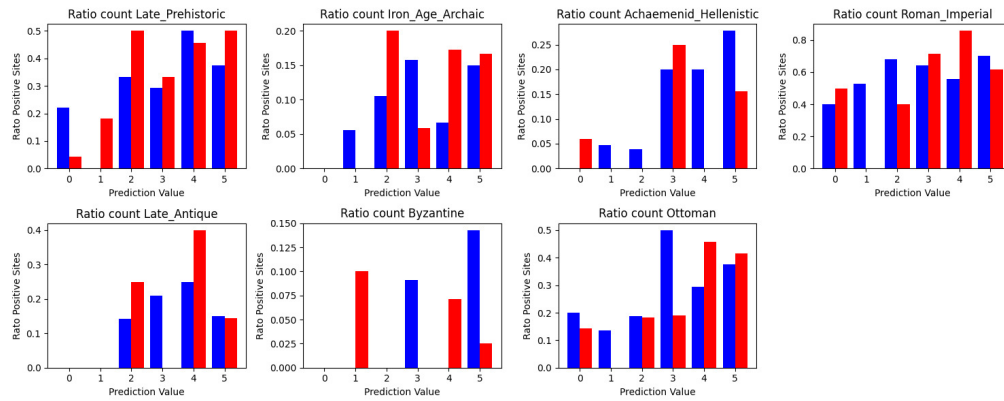


Figure 4.3: Bar graphs showing the ratio of positive plots to the total number of plots (y) for each predicted LAMAP (blue) and naive Bayes (red) class (x), for each time period. The height of each bar is effectively the empirical probability of finding something given a class.

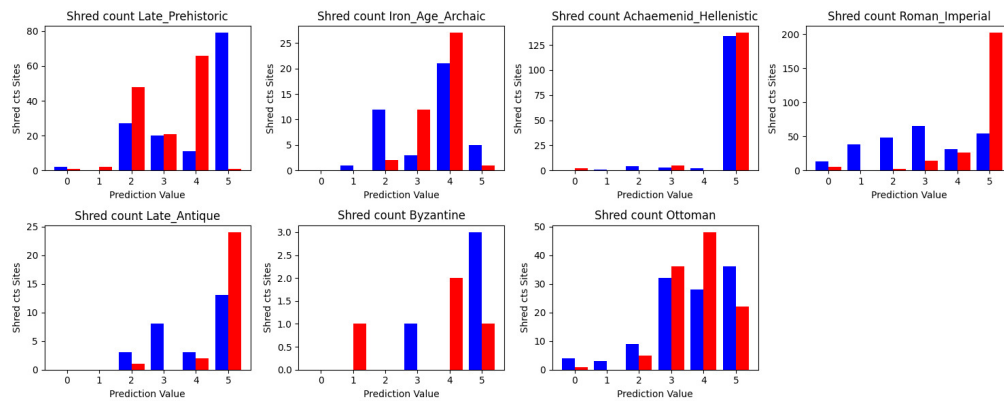


Figure 4.4: Plots showing the number of finds (y) per LAMAP (blue) and naive Bayes (red) class (x) per period

survey should be undertaken that also considers an equal distribution over classes for the naive Bayes method. Despite this, the naive Bayes method still shows potential as a method for archaeological predictive modelling in the survey area. If further research in a targeted study shows the same potential, naive Bayes has the potential to become a quick and easy-to-use method in archaeological predictive modelling.

### 4.3 Naive Bayes Binary site Prediction

The first preliminary test of naive Bayes in previous section shows that naive Bayes disposes of potential in modelling archaeological potential. In this section, cross validation will be used to understand the capabilities of naive Bayes to predict archaeological site occurrence. Both known sites and known non-site are used to construct this naive Bayes model. For a new data point consisting of local environment variables, the model gives a binary prediction. Output 1 indicates the model's belief that its input belongs to the archaeological site class while output 0 indicates belief in the input belonging to the non-site class. By using cross-validation, the model can be tested on the available data and thus no field study is needed. It is important to underline that this analysis does not provide insight into the ability of naive Bayes to predict archaeological potential as defined in section 2.3 but rather its performance in predicting site occurrence. However, it is still interesting as there is an intuitive link between modelling archaeological potential and prediction of site locations.

#### 4.3.1 Cross Validation Results

The metrics shown in Figure 4.5 are the result of the average over all time period of the average performance over the time period specific folds. In this context, the naive Bayes method according to the chosen metrics shows positive performance. The mean F1 score of 0.62 with a standard deviation of 0.12 shows the capacity of naive Bayes as a binary classification model for archaeological site prediction. Despite its strong underlying assumptions, naive Bayes is able to correctly classify positive unseen instance and minimise the number of false positives and false negatives. However, a previous high variation in the recall metric for positive instances results in a visible spread for the F1-score in the same binary class. The recall metric for the positive class reflects the ability to find all positive instances. In other words, the percentage of truly positive data points that were predicted as positive. The spread is the result of some inferior performance of the model in some individual time periods. The predictive ability of naive Bayes in individual time period will be illustrated next sections on the comparison between naive Bayes, LAMAP and the semantic segmentation method. It is notable, that the spread in variance, excluding a few outliers, is mainly located in the positive predictions. On average, the model seems to have more difficulty in correctly classifying all known-site instances compared to the non-site instances. To understand the entire predictive capability, the average of the metrics across the positive and negative instance validation is taken. In the test method adopted, naive Bayes shows a prediction equal to the ground truth in 66.67 of cases. An average of 64.08% instances of a class are identified as their ground truth class. Together, these metrics show an F1-score of 0.62 with a standard deviation of 0.07.

### 4.4 Comparative study

In this section, a comparison test follows between the LAMAP and the proposed machine learning techniques naive Bayes and semantic segmentation. To facilitate this comparison, an important

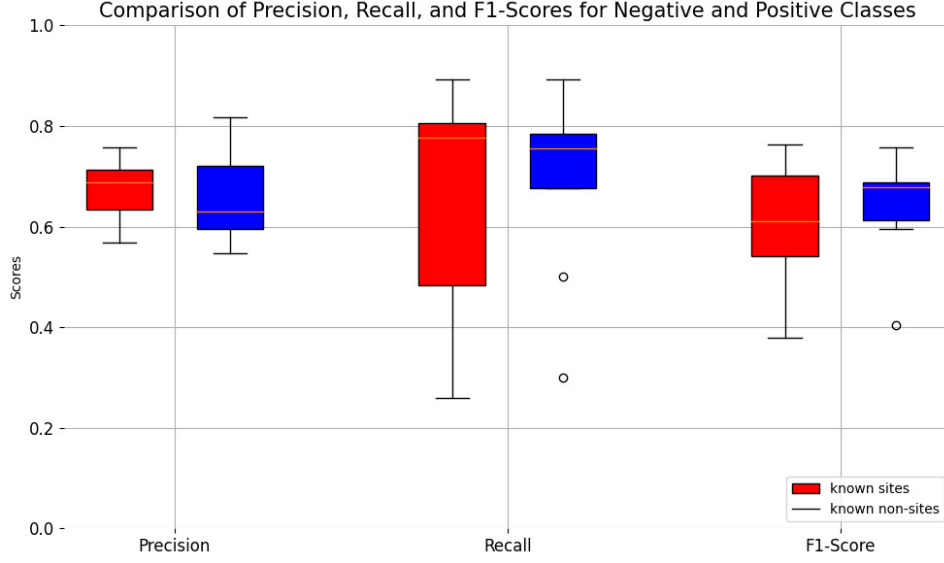


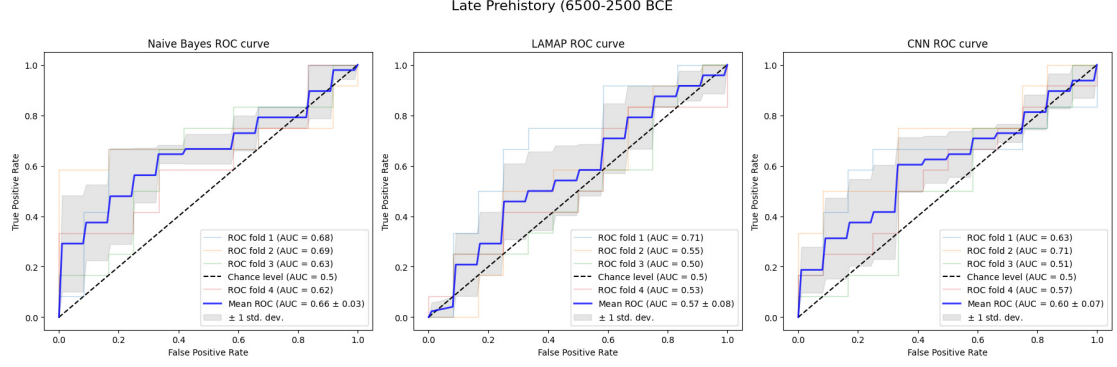
Figure 4.5: Boxplots showing a comparison of different cross-validation metrics on naive Bayes predictions averaged over all time periods.

assumption needs to be made. LAMAPs predictions are relative to their environment and only estimate the potential of a location. In contrast, naive Bayes and semantic segmentation effectively predict the probability of site occurrence. The difference in prediction type is nuanced but it should still be underlined that in the following comparison, it is assumed that LAMAPs type also has site prediction capabilities. The three models are compared by training and validation on the same 4-fold cross validation data. Training as well as test data consists of both known-sites and known non-site but only semantic segmentation also uses the known non-site class to model its hypothesis. In order to limit the LAMAPs computational needs, the number of neighbouring sites to be considered for this comparison is limited to 20. It is assumed that LAMAP's predictive capacity will not suffer significantly from this. This assumption stems from the small amount of available data and the advantage that only the 20 surrounding sites, which according to LAMAP provide the most valuable information, are taken into account. The purpose of this section is to understand the predictive value of the different output signals of the three models. Clearly, in this section, the output of naive Bayes and LAMAP is interpreted as the belief by which its input belongs to the known-site class. In both methods, their output signal informs about the predicted belief in its input being of the known-site class. Classically, semantic segmentation is used for categorical classification for which one categorical label, for which the raw prediction is highest, is assigned to each pixel. To compare this method with LAMAP and naive Bayes, the preceding process takes one step back and only the raw prediction for the known-site class is given as output. Note, however, that unlike naive Bayes and LAMAP, this prediction is influenced by experience on known non-site data. The test set consists of unseen known-site and known non-sites. A common way to convert a belief prediction to a binary prediction is to use a threshold of 0.5. Any arbitrary separation value can be chosen to balance error types discussed in section 4.1.2 and section 4.1.3. To compare the methods it makes no sense to define a single threshold for which predictions below are predicted non-site and predictions above are predicted sites. Due to the different interval of predictions in the three methods, this will give cluttered results. ROC-curves and AUC values discussed in section 4.1.2 provide the solution to this. The

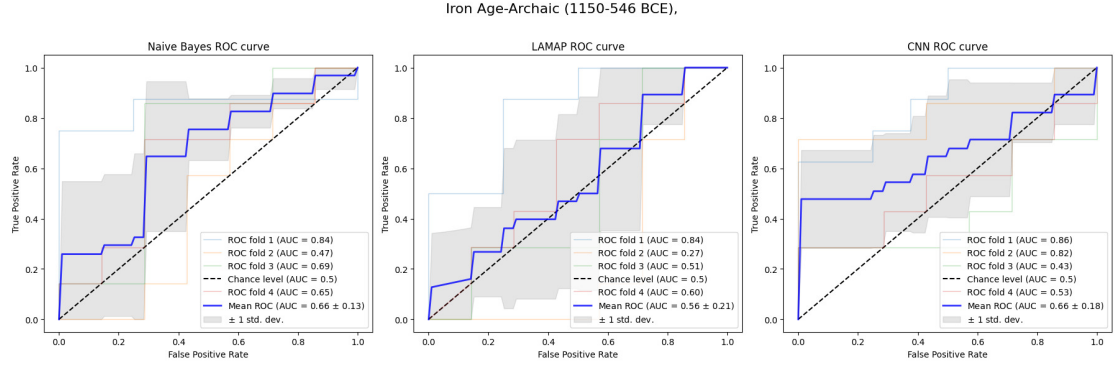
3 models are considered individually by plotting metrics for each splitting threshold individually. In this way, insight is gained into the predictive capabilities of the different models in predicting site occurrence.

#### 4.4.1 Comparison

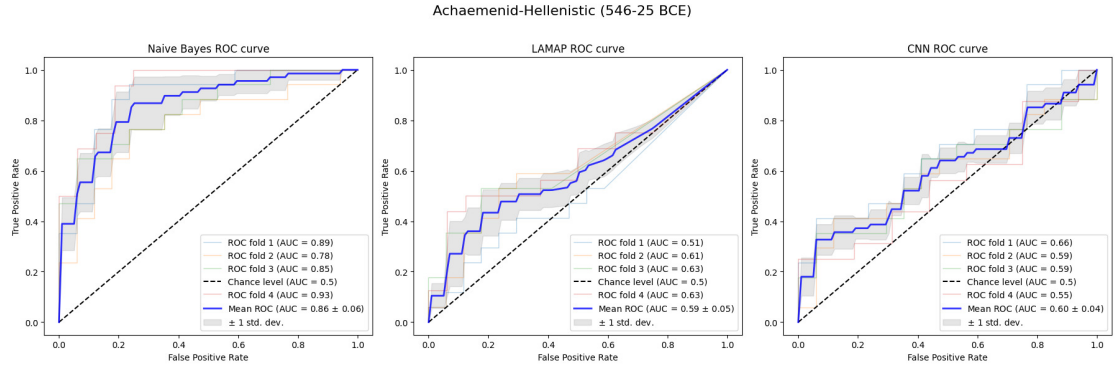
A first positive result is that for each time period, the models outperform random classification indicated by the dotted line on average. Remarkably, in the chosen comparison environment, the relatively simplistic naive Bayes has the best performance. Over all time period, naive Bayes yields to an AUC value of 0.73 on average with a standard deviation of 0.09. According to mean performance following naive Bayes is the semantic segmentation method with a mean AUC of 0.6 and a standard deviation of 0.08. LAMAP's result in this setting surprisingly show the worst results in comparison to the other methods. LAMAP scores a mean AUC of 0.56 with a standard deviation of 0.09. The convincingly better performance of naive Bayes versus LAMAP is surprising since LAMAP is intuitively thought to make more accurate local predictions via its local adaptivity. It should be emphasised that this comparison takes place within the chosen test method and so this is not a global claim about capabilities of the different models. The various nuances in this comparison are discussed further in the discussion in section 5. Compared to the other time periods, for the Iron Age-Archaic time period, there is high variability among the different folds. This is probably due to the small number (only 29) of known site data points and results in poor generalisation to unseen data. The relatively small variation in AUC values across the 4 folds for the other time periods and small overall variation suggests consistency of the models across different data subsets. This implies stable models which seem able to generalise across their training data.



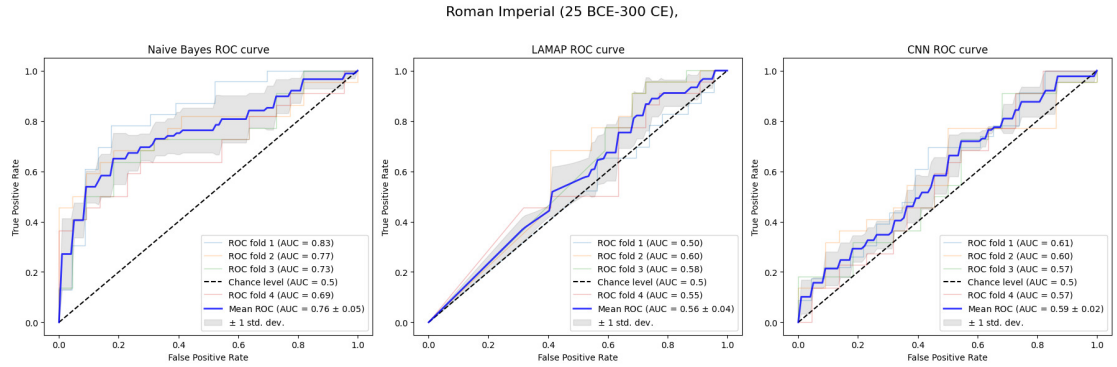
(a) ROC curves and AUC values for the Late Prehistoric time period.



(b) ROC curves and AUC values for the Iron Age-Archaic time period.

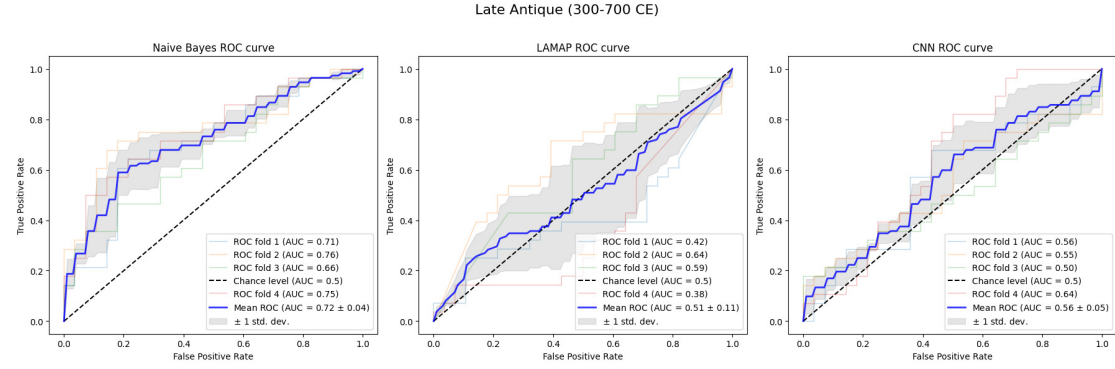


(c) ROC curves and AUC values for the Achaemenid-Hellenistic time period.

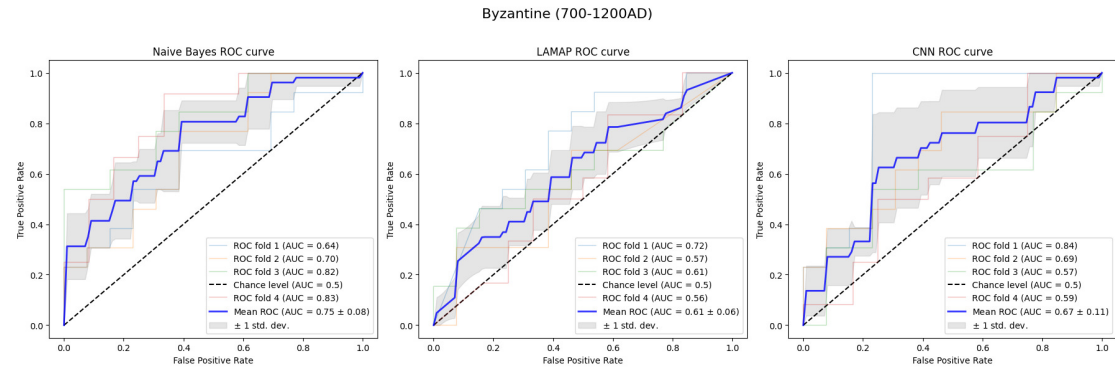


(d) ROC curves and AUC values for the Roman Imperial time period.

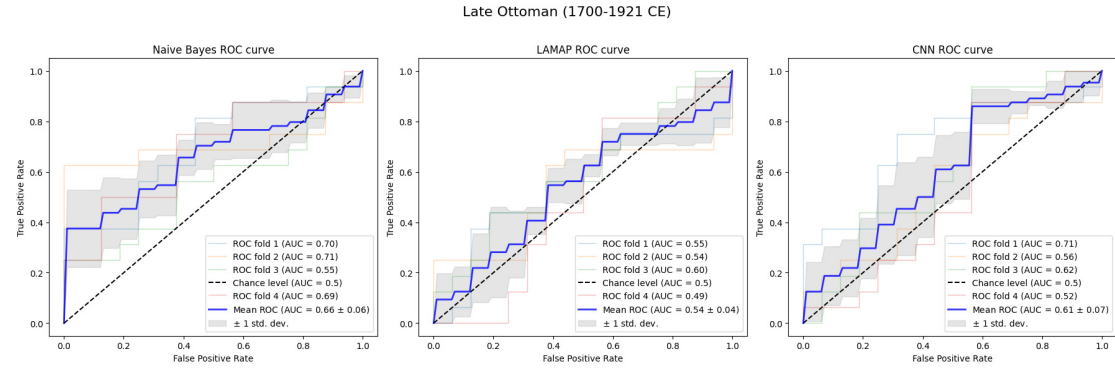
Figure 4.6: Plots showing obtained results for the first four time periods involved in this study.



(a) ROC curves and AUC values for the Late Antique time period.



(b) ROC curves and AUC values for the Byzantine time period.



(c) ROC curves and AUC values for the Late Ottoman time period.

Figure 4.7: Plots showing obtained results for the last three time periods involved in this study.

## Chapter 5

# Discussion

This dissertation introduced naive Bayes and semantic segmentation as possible tools for modelling archaeological potential. Naive Bayes addresses the high computational needs of LAMAP. Semantic segmentation was proposed to account also for complex geospatial dependencies of variables which in LAMAP and naive Bayes are assumed to be independent. The hypothesis for the experiments predicted semantic segmentation as the best performing model followed by LAMAP and naive Bayes as the least performing method. The superiority of semantic segmentation was expected based on the discussed successes of its applications to geospatial data. Naive Bayes was envisaged to be a helpful first tool in preliminary predictions of archaeological potential. The expectations suggested inferior performance against LAMAP because naive Bayes is not adaptive to its environment. This position adaptive property of LAMAP was proposed as crucial for accurate predictions in a geologically highly variable landscape. The proposed analyses probe the validity of these expectations. First, naive Bayes was directly compared with LAMAP using the survey study designed for LAMAPs validation in the Sagalassos survey area. In this study, through the construction of a regression model, a positive correlation was found between the potential predicted by naive Bayes and both the empirical probability of finding something and the number of artefacts found. While these results provide a positive indication about the potential of naive Bayes in this context, it is not possible to base general claims on this about the possible preference of naive Bayes over LAMAP. This is mainly due to the nature of the validation data which was collected according to an equal distribution across LAMAP prediction classes. In order to still probe general claims about the performance of naive Bayes, cross validation was performed as a 2nd study. On this, metrics such as recall, precision and the F1 score were used to provide insight into the capabilities of the probabilistic model for binary site, non-site classification. The study shows an average F1 score of about 0.62. In its setting, this result proves the potential of naive Bayes to classify known sites and known non-sites in its validation data using probability distributions. Following this, a 3rd study investigated the mutual ratio in predictive capacity of the two proposed machine learning method being naive Bayes and semantic segmentation versus the LAMAP method. The comparison was based on ROC curves and AUC values collected by running the three models over the same 4-fold cross-validation subsets. In this context, the hypothesis in performance, semantic segmentation > LAMAP > naive Bayes, was found to be incorrect. Naive Bayes showed convincingly superior while LAMAP showed the worst performance and semantic segmentation under-performed its expectations. However, again, it proves to be difficult to make general claims since some discrepancies were found between the different methods. Also, some concerns follow from the use of cross validation as a foundation for inter-comparison. These discrepancies and concerns are

presented in the following points.

### **On the Comparison between LAMAP and ML methods**

A single pixel consisting of a collection of values for each of the selected environmental factors at a given geographical position is the input to LAMAP. Using this method, as discussed in section 2.3.1, the probability by which a pixel like the input occurs in the available known-site dataset is calculated. This gives the archaeological potential of the input. High potential here means a proportionally high occurrence of a pixel like the input in the known-site set and low potential reflects the opposite. In contrast, the application of machine learning methods naive Bayes and semantic segmentation in the comparative study answers the question by which probability its input belongs to the class of archaeological sites. In the naive Bayes method, this is done explicitly by feature probability distributions, in semantic segmentation implicitly by a complex layered architecture. There is a subtle gap in the meaning of archaeological potential used between the machine learning methods and LAMAP. In this dissertation, however, the different methods were compared based on their potential to predict site locations. However, this binary prediction task is more in line with what naive Bayes and semantic segmentation are designed to do. This is an important nuance that needs to be taken into consideration for the interpretation of results. The assumption was also made that a limit of 20 on the number of sites to be considered would not have significant impact on its predictive capabilities compared to the other methods. However, it is not clear whether this assumption is valid. In its reliability study, LAMAP shows the strongest predictions for the Achaemenid-Hellenistic and Late Prehistoric periods. This is in line with the results obtained in this study where LAMAP performs 2nd and 3rd best in these periods, respectively. Also, both studies agree on Iron Age as the time period in which LAMAP shows the worst performance. However, a noticeable difference is observed for the Byzantine period. The applied experiment in this dissertation shows LAMAPs as most efficient for this period while for the same period in the reliability study the second lowest performance is found. Although the similarities indicate the usefulness of the proposed experiment, because of this significant difference, it cannot be proven equal to field study as a basis for the validation and comparison of APM models.

### **On Semantic Segmentation**

Naive Bayes and LAMAP have a great advantage to be deployed when only positive data labels are available. Both model their definition of archaeological potential based on available archaeological sites. Their predictions are not affected by examples of known non-site data. However, this contrasts with Semantic segmentation which needs both positive and negative data points so that it learns to maximise the space between predictions for the different instance classes in training examples. This is another example raising the question whether comparison between the methods is reasonable. Given semantic segmentation is trained over both positive and negative examples and subsequently tested on unseen data points from that same population, better performance was naturally expected compared to LAMAP and naive Bayes. However, the results show moderate performance. Semantic segmentation, except for application in the Iron Age-Archai period, shows similar results to LAMAP. Also, for each time period, the average AUC-values are positive which implies that this method has predictive value for each of these periods. The unmet expectations do not invalidate the potential of this method. Further research will have to show whether optimisations for the relatively simplistic semantic segmentation architecture used can fulfil them.



### On the Framework of Validation

The choice of cross validation as a framework for validating and comparing different models comes with some important nuances. The use of multiple training test distribution attempts to gain insight into model performance invariant to the data used. However, for LAMAP, due to its local adaptivity this poses a disadvantage versus naive Bayes and the Semantic segmentation method. In LAMAP, the evidence consisting of distributions of landscape variables for each site is weighted by its distance from the prediction. Archaeological sites closer to the prediction are given more weight than more distant sites. This introduces the belief that settlements in a particular region are more similar to settlements nearby which are subject to a similar geological environment. Making predictions more dependent on local evidence answers the difficulty of modelling in a geologically highly variable environment like Sagalassos. This makes LAMAPs predictions sensitive to their geospatial position versus the positions of available evidence. When no evidence is available in the vicinity of the prediction, sites at far distances, which according to the assumption have a lesser predictive value, are still weighted according to their distance. Intuitively, this raises the question of whether, in the case of little local evidence, it might be more desirable to average over the available data rather than weight them. The dependence of a prediction on the archaeological sites in its vicinity makes for important difficult comparability with other models. In this thesis, cross validation is used as a testing framework. A first obvious advantage of this is that it limits LAMAPs exponentially growing computational complexity since for each fold in the 4 fold validation uses 75% of the available data points. A disadvantage, however, is that although over the 4 different train/validation collections it is guaranteed that each site is used once in validation there is no consideration of spatial spacing between train and validation data points. In certain train/validation distribution, by chance, the sites in train may be distant from sites in the validation set. It should be noted that the main proposed idea, local adaptivity, in the locally-adaptive model of archaeological potential (LAMAP) is questionably efficient in this context. This problem is inherent in the use of incomplete data but is magnified in the use of the comparison method used. By not considering the available archaeological sites individually, both naive Bayes and semantic segmentation are to a lesser extent sensitive to local evidence. Especially deep machine learning architectures for accurate prediction given some features also benefit from training examples which are in the feature space near its input. It can be considered that the problem due to lack of local evidence, explicitly measured as distance in the spatial environment, in LAMAP is similar to the lack of local evidence, measured in distance in a complex feature space, for semantic segmentation. In naive Bayes, this concern is less explicit due to evidence modelled using assumed normal distributions as seen in section 2.4.

### On the Data Collection

We should also be critical of the data used in this comparative study carried out in section 4.4. All data on known archaeological sites and known non-archaeological sites discovered and documented on the basis of processes which, although attempts are made to avoid this, are subject to a bias. This can result in an uneven distribution of site data extraction across the study area. Of course, there is no need for predicative models if the full ground Truth is available which in this context means the availability of all archaeological sites in the survey area would mean. Also, the choice of defined properties of these sites such as local elevation, slope or distance from road networks and collection of their data is as well susceptible to bias by, for example, poorly performing GIS methods. The neglect of cultural variables in the construction of models can also result in overly environmentally deterministic outputs [81].

### 5.0.1 Positioning the results

Despite the foregoing critical aspects of the results, it is possible that naive Bayes can be very superior to the semantic segmentation method because it requires less data and also works on only positive data instances. Therefore, the fact that a sparse amount of data is a characteristic of archaeological site data and it consists predominantly of positive labels makes naive Bayes very attractive for archaeological predictive modelling. Cross validation breaks the LAMAPs local assumptions which also shows naive Bayes to be superior to this method in used experimental setup. There are research areas in which naive Bayes despite its simplicity shows better or at least similar performance compared to relatively much more complex neural networks [40, 69]. Perhaps modelling archaeological potential is also one such case.

### 5.0.2 Further Research

The potential contribution of semantic segmentation to APM remains unclear. A key prerequisite for effective semantic segmentation models is a large amount of data. These fundamental conditions were not met in the context of this thesis. However, a lack of labels is a common problem, especially in the use of remote sensing images, for which some solutions have already been described in the literature [34, 3]. Further research needs to investigate the implementation of these methods in the context of APM. A defining characteristic of APM is the common lack of negative data points. Positive unlabelled (PU) learning is the machine learning setting that specifically investigates how a model can learn using only positive and unlabelled instances. PU learning seems to offer an interesting starting point for future attempts to integrate machine learning into this field. Doubts were also written down about the comparison method used between naive Bayes, LAMAP and semantic segmentation. APM research lacks a standard benchmark on which models are compared. Further research is needed to address the discrepancies between models and the biased comparison method used in this dissertation.

## Chapter 6

# Conclusion

This dissertation explored to contribute to the field of archaeological predictive modelling which aims to model potential archaeological sites using landscape properties. Early attempts at modelling were based on rules established by experts. More data-oriented methods emerged after this which deduce rules for landscape variables showing positive correlation with the occurrence of a site. However, these methods are of limited use in highly variable geographical environments such as the Sagalassos area. LAMAP offers a solution to this through its spatial adaptability where more weight is given to known sites in the vicinity of the target prediction. Unfortunately, the operation facilitating this adaptability comes at a high computational cost of  $O(2^n)$  and is thus not scalable. Recent applications of APM use developments in machine learning to obtain robust models. This dissertation introduces two such machine learning models to the field. First, a rather simplistic machine learning method naive Bayes was proposed. Naive Bayes, like LAMAP, uses distributions of landscape variables to calculate probabilities. By combining these probabilities, the potential is predicted. An important difference between the two is that naive Bayes considers data on different archaeological sites not individually but as a whole. As a result, adaptability to the position to be predicted is not possible but huge gains are made in terms of necessary computational resources. Also, naive Bayes provides a uniform baseline model because it does not need any parameter selections as in LAMAP, for instance, variable intervals have to be chosen. There are however considerations limiting the applicability of LAMAP and naive Bayes. Both assume independence of landscape variables given their labels and thus cannot model interactions between them. Both produce predictions directly on their input variables, if these are not informative or dependent on other inputs they are still given equal weight and can lead to inaccurate predictions. These restrictions were addressed by introducing the semantic segmentation architecture. The experiments conducted sought to provide insight into the predictive capabilities of the different models. In the setting of the comparative study, naive Bayes showed convincingly as the best performing model. It became clear however that comparing these models is not evident and comes with many assumptions and nuances. Further work will have to define a framework in which different methods are comparable. This work acts as a first step in integrating neural networks in APM and can be used as a basis for further steps in this research. Semantic segmentation has the great advantage of having rich literature which can be used to optimise its application in APM. Finally, naive Bayes appears to be a potentially valuable addition to APM methods. It remains open whether the potential shown is also apparent outside the test setting used. To answer this a benchmark or, if this does not prove possible, fieldwork is needed. A lack of literature on the use of neural networks in APM suggests that the current AI revolution has not currently found its way into this field of research. This is emphasised by Yuval

Noah Harari's prediction archaeologists have a 97% chance of keeping their jobs in this revolution [30]. Archaeology is unfortunately only to a lesser extent subject of attempts to innovate with machine learning techniques. APM can undoubtedly take advantage of machine learning advances in other research areas but would also benefit from focused attention considering its characteristics.

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